Quantum accessible information and classical entropy inequalities

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Abstract

Computing accessible information for an ensemble of quantum states is a basic problem in quantum information theory. We show that the recently obtained optimality criterion (A.S. Holevo, Lobachevskii J. Math., **43**:7 (2022), 1646-1650), when applied to specific ensembles of states leads to nontrivial tight entropy inequalities that are discrete relatives of the famous log-Sobolev inequality. In this light, the hypothesis of globally information-optimal measurement for an ensemble of equiangular equiprobable states (quantum pyramids) (B.-G. Englert and J. Řeháček, J. Mod. Optics **57** N3 (2010) 218-226) is reconsidered and the corresponding entropy inequalities are proposed. Via the optimality criterion, this suggests also an approach to the proof of the conjectures concerning globally information-optimal observables for quantum pyramids.

Keywords and phrases: quantum state ensemble, accessible information, convex optimization, quantum pyramid, tight lower bound for Shannon entropy.

1 Introduction

The problem of accessible information – the maximal information that can be extracted from measurements over an ensemble of quantum states – is a basic problem in quantum information and communication theory, coming back to its origin, see e.g. [6], [16], [9], [11], and important in particular for applications in quantum cryptography. The problem was solved in some special (notably, symmetric) cases but in general remains open. In section (2) of this paper we describe a general approach following [7] which gives a criterion for optimality and ties up the problem with *entropy inequalities* – the tight lower bounds for the classical Shannon entropy. It turns out that the optimality criterion when applied to specific ensembles of states leads to new nontrivial entropy inequalities that are discrete "distant relatives" of the famous log-Sobolev inequality, as explained in section 3.

In this light, the conjecture of an information-optimal measurement for an ensemble of equiprobable equiangular states – *quantum pyramid*, formulated and numerically substantiated by Englert and Řeháček in [2] – is reconsidered in section 4 of the present paper. In the case of *acute* pyramids the corresponding parametric family of entropy inequalities (23) is proposed in subsections 4.1-4.3. The inequalities suggest the new tight lower bounds for the Shannon entropy of a discrete probability distribution as compared e.g. to the known bound from [13], theorem 3.11.

In the subsection 4.4 the case of *obtuse* pyramids is considered to get further insight and to formulate the hypotheses concerning the corresponding entropy inequalities (60), (68). Our approach allows also to derive the nonlinear equations (49) (resp. (65)) serving to determine the critical values of the parameter t of the optimal observable corresponding to transition from moderately to strongly acute (resp, obtuse) pyramids.

In the subsection 4.5 we derive the tight entropy inequality (72) implying the global optimality of the hypothetical observable from [2] (proposition 2 and corollary 3) in the case of *flat* pyramids. The findings in subsections 4.4, 4.5 also emphasize that for obtuse and flat pyramids in the dimensions $m \geq 7$ the sharp "square root" observable (SRM) outperforms the "unambiguous discrimination" observable (UDM) and tentatively is the optimal one.

The proposed entropy inequalities may be more convenient for numerical check than the direct optimization, however proving them analytically is a challenge. In section 5, the suggestions for proofs proposed by the second author as a response to the hypotheses (23), (60), (68) put forward by the first author, are presented. Via the optimality criterion of section 2, this suggests also an approach to the proof of the conjecture concerning globally information-optimal observables for quantum pyramids formulated and numerically substantiated in [2].

2 Accessible information: a criterion for op-

timality

Let $\mathcal{H}$ be a finite dimensional Hilbert space, $\mathfrak{S} = \mathfrak{S}(\mathcal{H})$ the convex set of quantum states (density operators on $\mathcal{H}$). *Ensemble* of quantum states is a collection $\mathcal{E} = \{\pi_j, \rho_j\}_{j=1,\dots,m}$ where $\rho_j \in \mathfrak{S}$, and $\pi = \{\pi_j\}$ is a probability distribution. Let $\mathcal{M} = \{M_k\}_{k=1,\dots,n}$ be an observable i.e. a collection of operators on $\mathcal{H}$ such that $M_k \geq 0$, $\sum_{k=1}^{n} M_k = I$ (unit operator). The joint distribution of "input" j and "output" k is then

$$p_{jk} = \pi_j \mathrm{Tr}\ \rho_j M_k$$

and the Shannon mutual information between j and k is (throughout the paper log denotes the binary logarithm)

$$I(\mathcal{E}, \mathcal{M}) = \sum_{j,k} p_{jk} \log \frac{p_{jk}}{\pi_j p_k},$$

with

$$p_k = \sum_j p_{jk} = \mathrm{Tr}\ \overline{\rho} M_k,$$

where $\overline{\rho} = \sum_j \pi_j \rho_j$ is the *average state* of the ensemble $\mathcal{E}$.

Accessible information of the ensemble is defined as

$$A(\mathcal{E}) = \sup_{\mathcal{M}} I(\mathcal{E}, \mathcal{M}). \tag{1}$$

It was shown [1] that the supremum is attained on an observable of the form $M_k = |\varphi_k\rangle\langle\varphi_k|$, $k = 1, \dots, n$ with $d \leq n \leq d^2$ and linearly independent M_k (if the matrices of ρ_j in some basis are real then the number of components is reduced to $\frac{d(d+1)}{2}$). In particular, the supremum can be replaced with the maximum.

From now on we assume that the *average state $\overline{\rho}$ is nondegenerate*. Introducing the *dual* ensemble $\mathcal{E}' = \{p_k, \sigma_k\}$ with

$$\sigma_k = p_k^{-1} \overline{\rho}^{1/2} M_k \overline{\rho}^{1/2} = |\phi_k\rangle\langle\phi_k|,$$

where

$$|\phi_k\rangle = p_k^{-1/2} \overline{\rho}^{1/2} |\varphi_k\rangle,$$

and observable $\mathcal{M}'$ with the components

$$M'_j = \overline{\rho}^{-1/2} (\pi_j \rho_j) \overline{\rho}^{-1/2} \tag{2}$$

we have the same average state

$$\overline{\sigma} = \sum_k p_k \sigma_k = \overline{\rho}$$

and the new expression for accessible information:

$$A(\mathcal{E}) = \max_{\mathcal{E}':\overline{\sigma}=\overline{\rho}} I(\mathcal{E}', \mathcal{M}') \tag{3}$$

with

$$\begin{aligned} I(\mathcal{E}', \mathcal{M}') &= I(\mathcal{E}, \mathcal{M}) = -\sum_j \pi_j \log \pi_j + \sum_{k,j} p_k p(j|k) \log p(j|k) \\ &= H(\pi) - \sum_k \mathrm{Tr}\ \left(p_k \sigma_k\right) K\left(\sigma_k\right), \end{aligned}$$

where $p(j|k) = \mathrm{Tr}\ \sigma_k M_j'$,

$$K(\sigma) = -\sum_j M_j' \log \mathrm{Tr}\ \sigma M_j'. \tag{4}$$

Therefore

$$A(\mathcal{E}) = H(\pi) - \min_{\mathcal{E}':\overline{\sigma}=\overline{\rho}} \sum_k \mathrm{Tr}\ \left(p_k \sigma_k\right) K\left(\sigma_k\right) \tag{5}$$

Remark 1. *On the other hand, (3) is also the value of $\overline{\rho}-$constrained classical capacity $C(\mathcal{M}', \overline{\rho})$ of the measurement channel corresponding to the observable $\mathcal{M}'$ [8]. If H is the Hamiltonian and E is the energy limit then the energy-constrained capacity is*

$$C(\mathcal{M}', E) = \max_{\overline{\rho}:\mathrm{Tr}\,\overline{\rho}H \leq E} C(\mathcal{M}', \overline{\rho})$$

and the full classical capacity is just $C(\mathcal{M}') = \max_{\overline{\rho}} C(\mathcal{M}', \overline{\rho})$.

Theorem 1. *The minimization problem*

$$\min_{\mathcal{E}':\overline{\sigma}=\overline{\rho}} \sum_k \mathrm{Tr}\ \left(p_k \sigma_k\right) K\left(\sigma_k\right) \tag{6}$$

has the dual problem

$$\max\left\{\mathrm{Tr}\,\overline{\rho}\Lambda : \Lambda^* = \Lambda,\ \Lambda \leq K(\sigma) \textit{ for all } \sigma \in \mathfrak{S}\right\}. \tag{7}$$

The following statements are equivalent:

(i) Λ_0 *is the solution of the problem (7); ensemble* $\mathcal{E}_0' = \{p_k^0, \sigma_k^0\}$ *with* $\sigma_k^0 = |\phi_k^0\rangle\langle\phi_k^0|$ *is the solution of the problem (6) (equivalently, observable* $M_k^0 = |\varphi_k^o\rangle\langle\varphi_k^0|$ *,* $k = 1, \dots, n$*, with*

$$|\varphi_k^0\rangle = \left(p_k^0\right)^{1/2} \overline{\rho}^{-1/2} |\phi_k^0\rangle,$$

maximizes the accessible information (1));

(ii) a. $\Lambda_0 \leq K(\sigma)$ *for all* $\sigma \in \mathfrak{S}$*;*

b. $[K(\sigma_k^0) - \Lambda_0]\, |\phi_k^0\rangle = 0$*,* $k = 1, \dots, n$*.*

(iii) a. *The entropy inequality*

$$-\sum_j \left\langle \psi | M_j' | \psi \right\rangle \log \left\langle \psi | M_j' | \psi \right\rangle \ \geq \left\langle \psi | \Lambda_0 | \psi \right\rangle, \tag{8}$$

holds for all unit vectors $\psi \in \mathcal{H}$*;*

b. *The unit vectors* $|\phi_k^0\rangle$ *turn (8) into equality:*

$$-\sum_j \left\langle \phi_k^0 | M_j' | \phi_k^0 \right\rangle \log \left\langle \phi_k^0 | M_j' | \phi_k^0 \right\rangle = \left\langle \phi_k^0 | \Lambda_0 | \phi_k^0 \right\rangle, \quad k = 1, \dots .n. \tag{9}$$

The solution of the dual problem (7) is unique on the support of the operator $\overline{\rho}$ *i.e.* $\Lambda_0 \overline{\rho} = \Lambda_0' \overline{\rho}$ *for any two solutions* Λ_0, Λ_0'*. In particular, if* $\overline{\rho}$ *is non-degenerate, then the solution of the dual problem is unique.*

According to (5), the value of the accessible information (5) is then

$$A(\mathcal{E}) = H(\pi) - \operatorname{Tr} \overline{\rho} \Lambda_0. \tag{10}$$

Proof. We introduce the POVM on $\mathfrak{S} = \mathfrak{S}(\mathcal{H})$ as

$$\Pi(d\sigma) = \sum_k (p_k \sigma_k)\, \delta_{\sigma_k}(d\sigma),$$

where $\delta_{\sigma_k}(d\sigma)$ is the $\delta-$measure concentrated in the point $\sigma_k \in \mathfrak{S}$. then the problem (6) can be written as[1]

$$\begin{array}{rcl} \operatorname{Tr} \displaystyle\int_{\mathfrak{S}} K(\sigma) \Pi(d\sigma) & \longrightarrow & \min \\ \Pi(d\sigma) & \geq & 0 \,, \\ \Pi(\mathfrak{S}) & = & \overline{\rho}, \end{array}$$

[1]We refer to [7] for rigorous general definition of the traced integral and formulations of the optimality conditions.

which is similar to the Bayes estimation problem [6].This is a convex programming problem for which the dual problem is (7) and the optimality conditions are (ii.a) and

$$\int_B [K(\sigma) - \Lambda_0]\, \Pi_0(d\sigma) = 0, \quad B \subset \mathfrak{S}(\mathcal{H}),$$

see [7] for detail. In terms of the optimal ensemble $\mathcal{E}_0' = \{p_k^0, \sigma_k^0\}$ the last condition can be rewritten as

$$\left[K(\sigma_k^0) - \Lambda_0\right]\, \sigma_k^0 = 0, \quad (p_k^0 > 0). \tag{11}$$

Note that by summing over k we obtain

$$\sum_k K(\sigma_k^0)\, p_k^0 \sigma_k^0 \ = \Lambda_0 \overline{\rho}, \tag{12}$$

where Λ_0 must be Hermitean operator. This implies that $\Lambda_0 \overline{\rho}$ is the same for all solutions of the dual problem.

By taking into account that $\sigma_k^0 = |\phi_k^0\rangle\langle\phi_k^0|$, we can rewrite (11) as

$$\left[K(\sigma_k^0) - \Lambda_0\right] |\phi_k^0\rangle = 0 \tag{13}$$

and (12) as

$$\sum_k K(\sigma_k^0)\, p_k^0 \left|\phi_k^0\rangle\langle\phi_k^0\right| = \Lambda_0 \overline{\rho}.$$

Taking into account (4), the condition (ii.a) becomes

$$-\sum_j \left\langle \psi | M_j' | \psi \right\rangle \log \mathrm{Tr}\ \sigma M_j' \geq \left\langle \psi | \Lambda_0 | \psi \right\rangle, \quad \psi \in \mathcal{H},$$

for all $\sigma \in \mathfrak{S}$. Without loss of generality we can assume that $\|\psi\| = 1$ here. Then for fixed ψ the minimal value of the lefthand side is

$$-\sum_j \left\langle \psi | M_j' | \psi \right\rangle \log \left\langle \psi | M_j' | \psi \right\rangle$$

since the difference is just the relative entropy of two probability distributions.

Thus we arrive to the entropy inequality (8) as an equivalent of the condition (ii.a). Further, by taking into account that $K(\sigma_k^0) - \Lambda_0 \geq 0$, the condition (ii.b) in the form (13) is the same as

$$\langle \phi_k^0 | K(\sigma_k^0) | \phi_k^0 \rangle = \langle \phi_k^0 | \Lambda_0 | \phi_k^0 \rangle$$

which via (4) means that the unit vectors $|\phi_k^0\rangle$, satisfying the constraint $\sum_k p_k^0 \, |\phi_k^0\rangle\langle\phi_k^0| = \overline{\rho}$, turn (8) into the equality (9). □

Especially important for us is the case of pure-states ensemble with $\rho_j = |\psi_j\rangle\langle\psi_j|$ where $|\psi_j\rangle; j = 1, \ldots, d$, are linearly independent vectors. In this case the resolution of the identity (2) is

$$M_j' = |e_j\rangle\langle e_j|,$$

where the vectors

$$|e_j\rangle = \sqrt{\pi_j}\overline{\rho}^{-1/2}|\psi_j\rangle; \quad j = 1, \ldots, d,$$

by necessity form an orthonormal basis (depending on the ensemble $\mathcal{E}$). Then the entropy inequality (8) takes the form

$$-\sum_{j=1}^{d} |z_j|^2 \log |z_j|^2 \geq \sum_{j=1}^{d} \lambda_{jk} \overline{z_j} z_k,$$

where $\lambda_{jk} = \langle e_j | \Lambda_0 | e_k \rangle$ and $z_j = \langle e_j | \psi \rangle$ are complex variables subject only to the constraint

$$\sum_{j=1}^{m} |z_j|^2 = 1. \tag{14}$$

The left-hand side is just the Shannon entropy of the p.d. $\left\{ |z_j|^2 ; j = 1, \ldots, d \right\}$, and such an inequality can be approached with the tools of classical analysis. We will need the following

Lemma 1. *Let λ_{jk} be real numbers. If the inequality*

$$\sum_{j=1}^{m} |z_j|^2 \log |z_j|^2 + \sum_{j=1}^{d} \lambda_{jk} \overline{z_j} z_k \leq 0 \tag{15}$$

holds for all real z_j satisfying the condition (14), then it holds for all complex z_j satisfying the same condition.

Proof. Let $z_j = x_j + iy_j$ for real x_j, y_j, then (14) amounts to

$$\sum_{j=1}^{m} x_j^2 + \sum_{j=1}^{m} y_j^2 = 1.$$

Denote $P = \sum_{j=1}^{m} x_j^2$, $1 - P = \sum_{j=1}^{m} y_j^2$. The function $f(\xi) = \xi \log \xi$ is convex, and λ_{jk} are real hence

$$\begin{aligned}
\sum_{j=1}^{m} |z_j|^2 \log |z_j|^2 + \sum_{j=1}^{d} \lambda_{jk} \overline{z_j} z_k &= \sum_{j=1}^{m} f\left(P\left(\frac{x_j}{\sqrt{P}}\right)^2 + (1-P)\left(\frac{y_j}{\sqrt{1-P}}\right)^2 \right) \\
&+ P \sum_{j=1}^{d} \lambda_{jk} x_j x_k + (1-P) \sum_{j=1}^{d} \lambda_{jk} y_j y_k \\
&\leq P \left[\sum_{j=1}^{m} f\left(\left(\frac{x_j}{\sqrt{P}}\right)^2 \right) + \sum_{j=1}^{d} \lambda_{jk} x_j x_k \right] \\
&+ (1-P) \left[\sum_{j=1}^{m} f\left(\left(\frac{y_j}{\sqrt{1-P}}\right)^2 \right) + \sum_{j=1}^{d} \lambda_{jk} y_j y_k \right] \leq 0,
\end{aligned}$$

because

$$\sum_{j=1}^{m} \left(\frac{x_j}{\sqrt{P}}\right)^2 = \sum_{j=1}^{m} \left(\frac{y_j}{\sqrt{1-P}}\right)^2 = 1,$$

and by assumption (15) holds for real z_j satisfying (14). $\square$

The optimality conditions can be useful in the following ways:

1. for a problem which has a hypothetical solution, the conditions can be used to verify its actual optimality. One inserts the conjectured p_k^0, σ_k^0 into (11) to compute Hermitean Λ_0 . Then the main difficulty will be proof of the entropy inequality (8). In the continuous variable version, this was our strategy in the solution of the related Gaussian optimizers conjecture for the capacity of Gaussian measurement channels [8], with the new generalizations of the famous log-Sobolev inequality as the entropy inequality.

2. for a problem which has a known solution, we get an entropy inequality (8) as a consequence of the optimality. Also one can obtain an alternative proof of the optimality if one can find an independent proof of

the entropy inequality. In the next section we illustrate this point with a simple example resulting in the yet nontrivial tight lower bounds for the binary Shannon entropy.

3. The uniqueness of the solution of the dual problem (assuming $\bar{\rho}$ nondegenerate) has the following useful consequence. Assume that the ensemble $\mathcal{E} = \{\pi_j, \rho_j\}_{j=1,\dots,m}$ is *symmetric* in that $\pi_j \equiv 1/m$ and there is a group of *-automorphisms $\{\alpha_g; g \in G\}$ of the algebra of operators on $\mathcal{H}$ acting transitively on the set of states $\{\rho_j\}_{j=1,\dots,m}$. Then Λ_0 is an invariant of this group: $\alpha_g^* [\Lambda_0] = \Lambda_0; g \in G$.

3 Example: two pure states

The case of ensemble of two states was treated previously in [12], [10]. For simplicity of demonstration we take two pure equiprobable states. Let $|e_0\rangle, |e_1\rangle$ be an orthonormal basis in two-dimensional Hilbert space. Consider ensemble $\mathcal{E} = \{\pi_\pm, \rho_\pm\}$, where $\pi_\pm = \frac{1}{2}, \rho_\pm = |\psi_\pm\rangle\langle\psi_\pm|$ with $|\psi_\pm\rangle = \cos\alpha/2|e_0\rangle \pm \sin\alpha/2|e_1\rangle$, $0 \le \alpha \le \pi$. As argued in [12] the accesible information

$$A(\mathcal{E}) = 1 - h\left(\frac{1+\sin\alpha}{2}\right) \tag{16}$$

is attained by the optimal observable $\mathcal{M} = \{M_+, M_-\}$ where $M_\pm = |e_\pm\rangle\langle e_\pm|$ and $|e_\pm\rangle = \frac{1}{\sqrt{2}}(|e_0\rangle \pm |e_1\rangle)$ is an orthonormal basis[2]. Here

$$h(t) = -t\log t - (1-t)\log(1-t)$$

is the classical binary entropy.

Let us apply theorem 1 giving a proof of the global optimality and, as a byproduct, a new entropy inequality. The average state of the ensemble $\mathcal{E}$ is

$$\overline{\rho} = \frac{1}{2}|\psi_+\rangle\langle\psi_+| + \frac{1}{2}|\psi_-\rangle\langle\psi_-| = \cos^2\alpha/2\,|e_0\rangle\langle e_0| + \sin^2\alpha/2\,|e_1\rangle\langle e_1|$$

with

$$\overline{\rho}^{1/2} = \cos\alpha/2\,|e_0\rangle\langle e_0| + \sin\alpha/2\,|e_1\rangle\langle e_1|$$

so that

$$\frac{1}{\sqrt{2}}\overline{\rho}^{-1/2}|\psi_\pm\rangle = |e_\pm\rangle$$

[2]Strictly speaking, the paper [12] contains the check of *local* optimality.

and

$$M'_{\pm}=\frac{1}{2}\overline{\rho}^{-1/2}\rho_{\pm}\overline{\rho}^{-1/2}=|e_{\pm}\rangle\langle e_{\pm}|\,, \tag{17}$$

which is also the conjectured optimal measurement. Therefore the conjectured optimal ensemble $\mathcal{E}'_0$ coincides with $\mathcal{E}$ because

$$\sigma^0_{\pm}=2\overline{\rho}^{1/2}\,|e_{\pm}\rangle\langle e_{\pm}|\,\overline{\rho}^{1/2}=|\psi_{\pm}\rangle\langle\psi_{\pm}|\,,\quad \pi^0_{\pm}=\frac{1}{2},$$

so that $|\phi^0_{\pm}\rangle=|\psi_{\pm}\rangle$. Let us check the optimality conditions for the ensemble $\mathcal{E}^0=\mathcal{E}$.

Introducing the important parameter

$$p=\frac{1+\sin\alpha}{2},\quad \frac{1}{2}\leq p\leq 1,$$

we have

$$\langle e_{\pm}|\psi_{\pm}\rangle=\sqrt{p},\quad \langle e_{\pm}|\psi_{\mp}\rangle=\sqrt{1-p}.$$

Incerting $\sigma^0_{\pm}=|\psi_{\pm}\rangle\langle\psi_{\pm}|$ into

$$K(\sigma)=-\sum_{\pm}|e_{\pm}\rangle\langle e_{\pm}|\log\langle e_{\pm}|\sigma|e_{\pm}\rangle$$

we obtain

$$\begin{aligned}K(\sigma^0_{+}) &= -\,|e_{+}\rangle\langle e_{+}|\log p-|e_{-}\rangle\langle e_{-}|\log(1-p),\\ K(\sigma^0_{-}) &= -\,|e_{+}\rangle\langle e_{+}|\log(1-p)-|e_{-}\rangle\langle e_{-}|\log p.\end{aligned}$$

The ensemble $\mathcal{E}$ is symmetric with respect to the reflection relative to the axis $|e_0\rangle$; according to the remark 3 at the end of previous section, Λ_0 should also obey this symmetry, hence

$$\Lambda_0=\lambda_0(p)\,|e_0\rangle\langle e_0|-\lambda_1(p)\,|e_1\rangle\langle e_1|\,. \tag{18}$$

Solving the conditions (ii.b)

$$\left[K(\sigma^0_{\pm})-\Lambda_0\right]\,|\phi^0_{\pm}\rangle=0$$

and taking into account that

$$\cos\alpha/2=\frac{1}{\sqrt{2}}\left(\sqrt{p}+\sqrt{1-p}\right),\quad \sin\alpha/2=\frac{1}{\sqrt{2}}\left(\sqrt{p}-\sqrt{1-p}\right),$$

results in

$$\lambda_0(p) = -\frac{\sqrt{p}\log p + \sqrt{1-p}\log(1-p)}{\sqrt{p}+\sqrt{1-p}},$$

$$\lambda_1(p) = \frac{\sqrt{p}\log p - \sqrt{1-p}\log(1-p)}{\sqrt{p}-\sqrt{1-p}},$$

provided $p > 1/2$.

Then the conditions (ii.b) are fulfilled while (ii.a) reduces to (8) which via (17), (18) amounts to the following parametric family of tight lower bounds for the classical binary entropy

Proposition 1. *For $p > 1/2$ and $0 \leq t \leq 1$ (in our case $t = |\langle e_+|\psi\rangle|^2, 1-t = |\langle e_-|\psi\rangle|^2$),*

$$h(t) \geq (\lambda_0(p) + \lambda_1(p))\sqrt{t(1-t)} - \frac{\lambda_1(p) - \lambda_0(p)}{2}, \tag{19}$$

where

$$\frac{\lambda_0(p) + \lambda_1(p)}{2} = \sqrt{p(1-p)}\frac{\log p - \log(1-p)}{2p-1} \equiv \mu_0(p),$$

$$\frac{\lambda_1(p) - \lambda_0(p)}{2} = \frac{p\log p - (1-p)\log(1-p)}{2p-1}.$$

The inequality (19) becomes equality for $t = p$ and $t = 1-p$.

In these relations $p > 1/2$; taking the limit $p \to 1/2$ the inequality (19) becomes

$$h(t) \geq 2\log e\sqrt{t(1-t)} - \log\frac{e}{2}. \tag{20}$$

The value $p = 1/2$ corresponds to the trivial case $\rho_+ = \rho_-$ with $A(\mathcal{E}) = 0$. It is remarkable, however, that in the limit $p \to 1/2$ we get the nontrivial entropy inequality (20). It is equivalent to an inequality which was a starting point in the original derivation of logarithmic Sobolev inequality by L. Gross [3].

Thus (19), (20) can be regarded as discrete "distant relatives" of the log-Sobolev inequality. In the paper [8] we elaborated continuous variable version of the optimality conditions and applied it in the bosonic Gaussian case; the corresponding entropy inequality turned out to be a generalization of the genuine log-Sobolev inequality.

Proof. For convenience of differentiation we pass to natural logarithms. Denote by $g(t,p)$ the righthand side of (19), then

$$\frac{d}{dt}h(t) = -\ln t + \ln(1-t), \tag{21}$$

$$\frac{d}{dt}g(t,p) = \mu_0(p)\frac{1-2t}{\sqrt{t\,(1-t)}}. \tag{22}$$

Taking into account the symmetry of the functions with respect to the point $t=1/2$, we can restrict to the interval $t\in[1/2,1]$. One can check that $h(p)=g(p,p)$ and $\frac{d}{dt}|_{t=p}h(t)=\frac{d}{dt}|_{t=p}g(t,p)$ which means that $h(t)$ is the envelop of the family $g(t,p)$; $p\geq 1/2$, with $t=p$ as the tangential points. By using (21), (22) we obtain the key relation

$$\frac{d}{dt}\left[h(t)-g(t,p)\right]=\left[\mu_0(p)-\mu_0(t)\right]\frac{2t-1}{\sqrt{t\,(1-t)}}.$$

The function

$$\mu_0(t)=\sqrt{t\,(1-t)}\frac{\ln t-\ln(1-t)}{2t-1}$$

is monotonously decreasing on $[1/2,1]$, which can be seen by showing its derivative is negative on $(1/2,1]$ (for this it is convenient to make a monotonic change of variable $u=\sqrt{t/(1-t)}$).

Thus $h(t)-g(t,p)=0$ for $t=p$, and $\frac{d}{dt}\left[h(t)-g(t,p)\right]$ has the same sign as $t-p$. Hence $h(t)-g(t,p)\geq 0$. □

Theorem 1 then implies the global optimality of the observable $\mathcal{M}=\{M_+,M_-\}$. Also

$$\mathrm{Tr}\,\overline{\rho}\Lambda_0=\lambda_0(p)\cos^2\alpha/2-\lambda_1(p)\sin^2\alpha/2=h(p)$$

and (10) implies $A(\mathcal{E})=1-h(p)$ in accordance with (16).

4 Multidimensional case

4.1 The lower bound for the Shannon entropy: formulation and discussion

The multidimensional generalization of (19) to be derived in what follows is:

Theorem 2. *For integer $m \geq 2$ and $p > \frac{m-1}{m}$*

$$-\sum_{j=1}^{m} t_j \log t_j \geq \mu_0(p) \left(\sum_{j=1}^{m} \sqrt{t_j} \right)^2 - \mu_1(p), \tag{23}$$

where $t_j \geq 0, \sum_{j=1}^{m} t_j = 1$ and

$$\mu_0(p) = \frac{\sqrt{\frac{p(1-p)}{m-1}} \left(\log p - \log \frac{1-p}{m-1} \right)}{\left(\sqrt{p} + (m-1) \sqrt{\frac{1-p}{m-1}} \right) \left(\sqrt{p} - \sqrt{\frac{1-p}{m-1}} \right)}, \tag{24}$$

$$\mu_1(p) = \frac{\sqrt{p} \log p - \sqrt{\frac{1-p}{m-1}} \log \frac{1-p}{m-1}}{\left(\sqrt{p} - \sqrt{\frac{1-p}{m-1}} \right)}. \tag{25}$$

This lower bound is tight: the graphs of the functions of $t_1, \ldots, t_m$ in the left and in the right sides of (23) are tangential at the points obtained by permutations of $t_1 = p, t_k = \frac{1-p}{m-1}, k = 2, \ldots, m$.

The inequality (23) arises in connection with the problem of accessible information for ensemble of m equiprobable equiangular pure states discussed in the next subsection. The suggestion for a proof of (23) will be given in section 5.1. Note that in the case $m = 2$ the inequality (23) reduces to (19).

For $m \geq 3$ the value $p = \frac{m-1}{m}$ is allowed giving the inequality

$$\begin{aligned} -\sum_{j=1}^{m} t_j \log t_j &\geq \frac{\log(m-1)}{m-2} \left(\sum_{j=1}^{m} \sqrt{t_j} \right)^2 - \left[\frac{m \log(m-1)}{m-2} - \log m \right] \\ &= \log m - \frac{m \log(m-1)}{m-2} \left[1 - B(P; P_U)^2 \right], \end{aligned} \tag{26}$$

where

$$B(P; P_U) = \sum_{j=1}^{m} \sqrt{t_j / m}$$

is the Bhattacharyya coefficient between the probability distribution $P = (t_1, \ldots, t_m)$ and the uniform distribution $P_U = (1/m, \ldots, 1/m)$.

The inequality (26) then gives a lower bound for the Shannon entropy in terms of a "distance" between the distribution P and the uniform distribution

P_U, related to the Bhattacharyya coefficient. It can be compared with the lower bound of theorem 3.11 in [13]:

$$-\sum_{j=1}^{m} t_j \log t_j \geq \log m - \frac{m \log m}{2\,(m-1)} D(P; P_U), \tag{27}$$

where $D(P;P_U) = \sum_{j=1}^{m} |t_j - 1/m|$ is the total variation distance. There is well-known relation for any two distributions P, P':

$$2\,(1 - B(P;P')) \leq D(P;P') \leq 2\sqrt{1 - B(P;P')^2},$$

which allows to obtain from (26) a degraded version of (27) and vice versa. For P close to P_U, the bound (26) is substantially better than (27) while close to degenerate distributions with $B(P;P_U) \approx \sqrt{1/m}$ the right-hand side of (26) becomes negative which never happens for (27).

Similarly to (27), the inequality (26) (as well as (23)) can be rewritten as a tight upper bound for the classical relative entropy (the Kullback-Leibler divergence)

$$H(P||P_U) \leq \frac{m \log (m-1)}{m-2} \left[1 - B(P;P_U)^2\right]. \tag{28}$$

4.2 The conjecture: "quantum pyramids"

Let $|e_j\rangle$, $j = 1, \ldots, m$, be an orthonormal basis in $m-$dimensional Hilbert space.

$$|e_0\rangle = \frac{1}{\sqrt{m}} \sum_{j=1}^{m} |e_j\rangle, \quad \langle e_0|e_j\rangle \equiv \frac{1}{\sqrt{m}}.$$

Following Englert and Řeháček [2] we consider the system of equiangular vectors – the *quantum pyramid*:

$$|\psi_j\rangle = \sqrt{mr_1}|e_j\rangle + (\sqrt{r_0} - \sqrt{r_1})\,|e_0\rangle, \quad j = 1, \ldots, m, \tag{29}$$

$$r_0,\, r_1 \geq 0, \quad r_0 + (m-1)r_1 = 1. \tag{30}$$

Alternative parametrization was given in [4], section VI.2, where this configuration was related to "orthogonal signals" in communication theory:

$$|\psi_j\rangle = a|e_j\rangle + b\sum_{k\neq j} |e_k\rangle, \quad j = 1, \ldots, m, \tag{31}$$

$$a^2 + (m-1)b^2 = 1, \quad a \geq \frac{1}{\sqrt{m}}.$$

The relation between the two parametrizations is

$$a = \sqrt{mr_1} + \frac{\sqrt{r_0} - \sqrt{r_1}}{\sqrt{m}}, \quad b = \frac{\sqrt{r_0} - \sqrt{r_1}}{\sqrt{m}}, \tag{32}$$

$$1 - r_0 = (m-1)r_1 = \frac{m-1}{m}(a-b)^2. \tag{33}$$

The cosine of angle between any two different edges is

$$\xi \equiv \langle \psi_j | \psi_k \rangle = r_0 - r_1 = 2ab + (m-2)b^2\,, \qquad j \neq k, \tag{34}$$

thus the case $r_0 > r_1\,(b > 0)$ corresponds to the *acute* $(0 < \xi < 1)$, $r_0 = r_1\,(b = 0)$ – to the *orthogonal* $(\xi = 0)$, and $r_0 < r_1\,(b < 0)$ – to the *obtuse* $\left(-\frac{1}{(m-1)} < \xi < 0\right)$ pyramids. The extreme cases are $r_1 = 0$ $\left(a = b = \sqrt{\frac{1}{m}}\right)$ – the *ort* $|\psi_j\rangle \equiv |e_0\rangle$ $(\xi = 1)$, and $r_0 = 0$ $\left(a = \sqrt{\frac{m-1}{m}}, b = -\sqrt{\frac{1}{m(m-1)}}\right)$ – the *flat* $\left(\xi = -\frac{1}{(m-1)}\right)$ pyramid.

We will consider the ensemble $\mathcal{E} = \{\pi_j, \rho_j\}_{j=1,\dots,m}$ with $\pi_j = \frac{1}{m}$, $\rho_j = |\psi_j\rangle\langle\psi_j|$ with the average state

$$\bar{\rho} \equiv \frac{1}{m} \sum_{j=1}^{m} |\psi_j\rangle\langle\psi_j| = r_1 I + (r_0 - r_1)\,|e_0\rangle\langle e_0|. \tag{35}$$

Remark 2. *Operators of the form $c_1 I + c_0 |e_0\rangle\langle e_0|$ constitute Abelian algebra of operators invariant under the group of all *-automorphisms generated by the orthogonal transformations leaving the quantum pyramid invariant.*

One can then easily check that

$$\bar{\rho}^{1/2} = \sqrt{r_1} I + (\sqrt{r_0} - \sqrt{r_1})\,|e_0\rangle\langle e_0| \tag{36}$$

and

$$|\psi_j\rangle = \sqrt{m}\bar{\rho}^{1/2}|e_j\rangle. \tag{37}$$

In what follows we consider the solution for the accessible information $A(\mathcal{E})$ and its optimizing observable conjectured and supported numerically in [2]. In the rest of this subsection and in the next subsection we consider acute pyramids for which $r_0 \geq r_1$ or, equivalently, $b \geq 0$. According to [2] the conjectured optimal observable belongs to the family depending on the parameter $0 \leq t \leq 1$ and has the form $\mathcal{M} = \{M_k; k = 0, 1, \ldots, m\}$ where $M_k = |\tilde{e}_k\rangle\langle\tilde{e}_k|$

$$|\tilde{e}_k\rangle = |e_k\rangle + \frac{t-1}{\sqrt{m}}|e_0\rangle; \quad k = 1, \ldots, m, \tag{38}$$

$$|\tilde{e}_0\rangle = \sqrt{1-t^2}|e_0\rangle.$$

The optimal value of the parameter is

$$t_a(m) = \begin{cases} 1, & r_0 \leq \frac{4(m-1)}{m^2} \\ \frac{2(m-1)}{m-2}\sqrt{\frac{r_1}{r_0}}, & r_0 > \frac{4(m-1)}{m^2} \end{cases}, \tag{39}$$

giving the accessible information (eq. (36) in [2])

$$A(\mathcal{E}) = \begin{cases} I^{(SRM)}, & r_0 \leq \frac{4(m-1)}{m^2} \\ \frac{m(1-r_0)}{m-2}\log(m-1), & r_0 > \frac{4(m-1)}{m^2} \end{cases}, \tag{40}$$

where

$$\begin{aligned} I^{(SRM)} &= \frac{1}{m}\left(\sqrt{r_0} + (m-1)\sqrt{r_1}\right)^2 \log\left(\sqrt{r_0} + (m-1)\sqrt{r_1}\right)^2 \\ &+ \frac{m-1}{m}\left(\sqrt{r_0} - \sqrt{r_1}\right)^2 \log\left(\sqrt{r_0} - \sqrt{r_1}\right)^2 \\ &= \log m + a^2 \log a^2 + (m-1)\, b^2 \log b^2. \end{aligned} \tag{41}$$

Note that in the case $t_a(m) = 1$ the conjectured information-optimal observable $M_k = |e_k\rangle\langle e_k|$, $k = 1, \ldots, m$, is optimal for the Bayes discrimination problem for the ensemble $\mathcal{E}$ [4], sec. VI.2. It is called square-root measurement (SRM) since the system $\{|e_k\rangle\}$ is obtained from $\{|\psi_k\rangle\}$ by applying inverse square root of its Gram matrix, cf. (37).

Remark 3. *An upper bound and a benchmark for $A(\mathcal{E})$ is the classical capacity of the c-q channel $j \to |\psi_j\rangle\langle\psi_j|$:*

$$C = \max_{\pi} H\left(\sum_{j=1}^{m} \pi_j |\psi_j\rangle\langle\psi_j|\right) = H(\bar{\rho}).$$

The last equality follows from the symmetry of the quantum pyramid and concavity of the von Neumann entropy. According to (35) the density operator $\bar{\rho}$ has the eigenvalues r_0, r_1, the last of multiplicity $m-1$, hence

$$\begin{aligned} C &= -r_0 \log r_0 - (m-1) r_1 \log r_1 \\ &= -\left(1 - \frac{m-1}{m}(a-b)^2\right) \log \left(1 - \frac{m-1}{m}(a-b)^2\right) \\ &- \frac{m-1}{m}(a-b)^2 \log \frac{(a-b)^2}{m}, \end{aligned}$$

where in the second equality we used (33).

Unlike the case of accessible information, this expression is the same for all values of the parameters, for acute, obtuse and flat pyramids. The ratio $C/A(\mathcal{E})$ gives the gain due to the measurement entanglement at the channel's output. In particular, it tends to the infinity as the pyramid collapses to ort (weak signal) and equals to 1 for rectangular pyramid (ideal channel).

4.3 Checking the optimality conditions: the case of acute pyramids

Let us apply our approach of section 2. By using (37) the dual observable $\mathcal{M}'$ has the components

$$M'_j = \overline{\rho}^{-1/2} \left(\frac{1}{m} |\psi_j\rangle\langle\psi_j| \right) \overline{\rho}^{-1/2} = |e_j\rangle\langle e_j|, \tag{42}$$

hence

$$K(\sigma) = -\sum_j |e_j\rangle\langle e_j| \log \langle e_j|\sigma|e_j\rangle.$$

From (38), (36) we obtain the *dual* ensemble $\mathcal{E}' = \{p_k, \sigma_k\}$ with

$$\sigma_k = p_k^{-1} |\tilde{\phi}_k\rangle\langle\tilde{\phi}_k|, \quad p_k = \langle\tilde{\phi}_k|\tilde{\phi}_k\rangle$$

where

$$\begin{aligned} |\tilde{\phi}_k\rangle &= \overline{\rho}^{1/2} |\tilde{e}_k\rangle = \sqrt{r_1} |e_k\rangle + \frac{\sqrt{r_0}t - \sqrt{r_1}}{\sqrt{m}} |e_0\rangle; \\ p_k &= r_1 + \frac{r_0 t^2 - r_1}{m}; \quad k = 1, \dots, m; \end{aligned}$$

$$|\tilde{\phi}_0\rangle = \overline{\rho}^{1/2}|\tilde{e}_0\rangle = \sqrt{r_0\left(1-t^2\right)}|e_0\rangle; \quad p_0 = r_0\left(1-t^2\right).$$

Substitution gives

$$K\left(\sigma_k\right) = -\sum_j |e_j\rangle\langle e_j| \, \log p_k^{-1} \, \left|\langle e_j|\tilde{\phi}_k\rangle\right|^2,$$

hence

$$K\left(\sigma_k\right) = \begin{cases} -\sum_j |e_j\rangle\langle e_j| \, \log p_k^{-1} \, \left|\sqrt{r_1}\delta_{jk} + \frac{\sqrt{r_0}t-\sqrt{r_1}}{m}\right|^2, & k \geq 1 \\ -\sum_j |e_j\rangle\langle e_j| \, \log \frac{1}{m} = (\log m)\, I, & k = 0 \end{cases}.$$

Having in mind checking the optimality condition (ii.b) we compute

$$\begin{aligned} K\left(\sigma_k\right)|\tilde{\phi}_k\rangle &= -\sum_j |e_j\rangle \left(\sqrt{r_1}\delta_{jk} + \frac{\sqrt{r_0}t-\sqrt{r_1}}{m}\right) \log \frac{\left|\sqrt{r_1}\delta_{jk} + \frac{\sqrt{r_0}t-\sqrt{r_1}}{m}\right|^2}{r_1 + \frac{r_0t^2-r_1}{m}} \\ &= A(t)|e_k\rangle + B(t)\sqrt{m}|e_0\rangle; \quad k = 1,\ldots,m, \end{aligned}$$

where

$$A(t) = -\left(\sqrt{r_1} + \frac{\sqrt{r_0}t-\sqrt{r_1}}{m}\right) \log \frac{\left|\sqrt{r_1} + \frac{\sqrt{r_0}t-\sqrt{r_1}}{m}\right|^2}{r_1 + \frac{r_0t^2-r_1}{m}} - B(t), \tag{43}$$

$$B(t) = -\left(\frac{\sqrt{r_0}t-\sqrt{r_1}}{m}\right) \log \frac{\left|\frac{\sqrt{r_0}t-\sqrt{r_1}}{m}\right|^2}{r_1 + \frac{r_0t^2-r_1}{m}} \tag{44}$$

and

$$K\left(\sigma_0\right)|\tilde{\phi}_0\rangle = \sqrt{r_0\left(1-t^2\right)} \log m \, |e_0\rangle.$$

Motivated by the symmetry of the problem (discrete rotations around the axis $|e_0\rangle$) and by the remark 2 we look for Λ_0 in the form

$$\Lambda_0 = c_1 I + c_0|e_0\rangle\langle e_0|. \tag{45}$$

Then

$$\begin{aligned} \Lambda_0|\tilde{\phi}_k\rangle &= c_1\sqrt{r_1}|e_k\rangle + \frac{\left(c_0+c_1\right)\sqrt{r_0}t - c_1\sqrt{r_1}}{\sqrt{m}}|e_0\rangle; \; k \geq 1, \\ \Lambda_0|\tilde{\phi}_0\rangle &= \left(c_0+c_1\right)\sqrt{r_0\left(1-t^2\right)}|e_0\rangle. \end{aligned}$$

Comparing coefficients in the condition (ii.b): $\Lambda_0|\tilde{\phi}_k\rangle = K(\sigma_k)|\tilde{\phi}_k\rangle$, we obtain the system of equations

$$\begin{aligned} c_1\sqrt{r_1} &= A(t), \\ (c_0+c_1)\sqrt{r_0}t - c_1\sqrt{r_1} &= mB(t), \\ (c_0+c_1)\sqrt{1-t^2} &= \log m\sqrt{1-t^2}. \end{aligned} \tag{46}$$

Two cases are possible: $t = 1$ and $t < 1$. In the first case the third equation disappears while the first two turn into

$$\begin{aligned} c_1\sqrt{r_1} &= \frac{b\log b^2 - a\log a^2}{\sqrt{m}}, \\ (c_0+c_1)\sqrt{r_0} - c_1\sqrt{r_1} &= -\sqrt{m}b\log b^2, \end{aligned}$$

where we have used (43), (44) and (32). Taking into account the relations (32), we have $\sqrt{mr_1} = a - b$, and the solution is

$$\begin{aligned} c_1 &= \frac{b\log b^2 - a\log a^2}{a-b}, \\ c_0 &= m\frac{ab\left(\log a^2 - \log b^2\right)}{(a-b)\left[a+(m-1)b\right]}. \end{aligned} \tag{47}$$

In the case $t < 1$ the third equation in the system (46) reduces to $c_0 + c_1 = \log m$. Eliminating c_0, c_1 one obtains the equation for t :

$$A(t) + mB(t) = (\log m)\sqrt{r_0}t. \tag{48}$$

Substituting the expressions (43), (44) for $A(t), B(t)$ and introducing the parameter

$$\tau = \sqrt{\frac{r_0}{r_1}}t,$$

we can reduce (48) to the form

$$\tau\log\left(m-1+\tau^2\right) = \frac{m-1+\tau}{m}\log\left(m-1+\tau\right)^2 + \frac{m-1}{m}\left(\tau-1\right)\log\left(\tau-1\right)^2. \tag{49}$$

Denoting $s = \tau^{-1}$ equation (49) can be transformed into

$$\begin{aligned} f(s) &\equiv \ln\left[1+(m-1)s^2\right] \\ &- 2\frac{1+(m-1)s}{m}\ln\left[1+(m-1)s\right] - 2\frac{m-1}{m}(1-s)\ln|1-s| = 0. \end{aligned}$$

At the point $s = 0$ (corresponding to $\tau \to \infty$) this function and its first and second derivatives vanish, while $f'''(s) = 2\,(m-1)\,(m-2) > 0$ hence f is increasing and positive for small s. But $f(1) = -\ln m < 0$, hence there is a (unique) root in the interval $(0, 1]$, namely $s = \frac{m-2}{2(m-1)}$. There are no roots for $s > 1$ because the function is decreasing.

Thus the value $\tau_a(m) = \frac{2(m-1)}{m-2}$ suggested by (39) is the unique positive solution of (48). Substituting this value in the first equation of (46) and using the third equation we obtain

$$c_1 = \log m - \frac{m \log (m-1)}{m-2}, \quad c_0 = \frac{m \log (m-1)}{m-2}, \tag{50}$$

the values which do not depend on the parameters of the pyramid r_0, r_1. The criterion for the applicability of this second case is

$$t_a(m) \equiv \sqrt{\frac{r_1}{r_0}}\tau_a(m) = \sqrt{\frac{r_1}{r_0}}\frac{2(m-1)}{m-2} < 1$$

which, via (30), is equivalent to $r_0 > \frac{4(m-1)}{m^2}$, and, via (32), to $a^2 < \frac{m-1}{m}$.

Correspondingly, the condition for applicability of the solution (47) is $a^2 \geq \frac{m-1}{m}$. By introducing the parameter p such that

$$p = a^2, \quad \frac{1-p}{m-1} = b^2, \tag{51}$$

we have $b = \sqrt{\frac{1-p}{m-1}}$, leading to

$$\Lambda_0 = c_1 I + c_0 |e_0\rangle\langle e_0| = -\mu_1(p) I + \mu_0(p) m |e_0\rangle\langle e_0|$$

with $\mu_1(p), \mu_0(p)$ given by (25), (24) under the condition that $p \geq \frac{m-1}{m}$. Introducing complex numbers $z_j = \langle e_j|\psi\rangle$, so that $\sum_{j=1}^m |z_j|^2 = 1$, the optimality condition (ii.a) amounts to the inequality

$$-\sum_{j=1}^m |z_j|^2 \log |z_j|^2 \geq \mu_0(p) \left| \sum_{j=1}^m z_j \right|^2 - \mu_1(p).$$

Since $\mu_0(p) \geq 0$, this is equivalent to (23) where $t_j = |z_j|^2$.

Remarkably, inserting the limiting value $p = \frac{m-1}{m}$ we obtain for $m > 2$

$$\mu_1 = \frac{m \log (m-1)}{m-2} - \log m, \quad \mu_0 = \frac{\log (m-1)}{m-2},$$

which correspond to the values c_0, c_1, given by (50) and obtained by solving (46) in the case $t_a(m) < 1$.

Finally, from (45) and (35) $\operatorname{Tr} \Lambda_0 \bar{\rho} = c_1 + c_0 r_0$ so that

$$A(\mathcal{E}) = \log m - c_1 - c_0 r_0. \tag{52}$$

Thus we have the following corollary to the theorem 2:

Corollary 1. *For ensemble of acute pyramids the information-optimal observable is given by the equations (38), (39) as conjectured in [2]. The accessible information is given by the expressions*

$$A(\mathcal{E}) = \begin{cases} \log m + p \log p + (1-p) \log \frac{1-p}{m-1}, & p \in \left[\frac{m-1}{m}, 1\right] \\ \frac{m-1}{m-2} \log(m-1) \left(\sqrt{p} - \sqrt{\frac{1-p}{m-1}}\right)^2, & p \in \left[\frac{1}{m}, \frac{m-1}{m}\right] \end{cases} \tag{53}$$

which agree with (40).

Proof. For accessible information we use the relation (52). A routine calculation using (47), (32), (51) gives (41) which is the first line in (53) for moderately acute pyramids. For strongly acute pyramid, using (50), one gets

$$A(\mathcal{E}) = \frac{m\,(1-r_0)}{m-2} \log(m-1),$$

which is the second line, taking into account (33). □

4.4 The case of obtuse pyramids

When $r_0 < r_1$ ($b < 0$) the pyramid is obtuse, with extreme flat case $r_0 = 0$ when all the state vectors $|\psi_j\rangle$ lie in the hyperplane orthogonal to $|e_0\rangle$. The hypothetical optimal observable is then drastically different. To describe it let us first briefly survey the flat pyramid in the case $m = 3$ where $|\psi_j\rangle; j = 1, 2, 3$, are essentially the "trine" of equiangular unit vectors in two-dimensional real plane, cf. [5]. Then the optimal observable is $\mathcal{M} = \{M_k; k = 1, 2, 3\}$, where $M_k = \frac{2}{3} |\psi_k^\perp\rangle\langle\psi_k^\perp|$, and $|\psi_k^\perp\rangle$ are the unit vectors in the plane such that $\langle\psi_k^\perp|\psi_k\rangle = 0; k = 1, 2, 3$ [14]. In other words, $\mathcal{M}$ is "unambiguous discrimination" measurement (MUD) for the states $\rho_j = |\psi_j\rangle\langle\psi_j|$, $j = 1, 2, 3$. In the section 4.5 we perform the analysis based on our optimality criterion showing that the conditions (ii) of the theorem are fulfilled with $\Lambda_0 = I$, with the condition (ii.a) equivalent to the entropy inequality

$$-\sum_{j=1}^{3} t_j \log t_j \geq 1; \quad t_j = \frac{2}{3}\cos^2\left(\alpha + \frac{2\pi j}{3}\right), \tag{54}$$

which is saturated for $\alpha = \pi/2 + \pi k$.

Most interesting is the case of "nearly flat" trine when $0 < r_0 < \bar{r}_0 \approx 0,0614$. Then, as shown by P. Shor in [15], the conjectured optimal observable has 6 outcomes and is a statistical mixture of the MUD described above and the SRM $M_k = |e_k\rangle\langle e_k|$, $k = 1, 2, 3$. The SRM becomes optimal for obtuse pyramids with $\bar{r}_0 \leq r_0 \leq r_1 = 1/3$.

A generalization of this picture to the case $m \geq 3$ based on numerical studies was suggested and elaborated in [2]. Let us apply our optimality criterion to derive the corresponding entropy inequalities and to obtain additional insight. We consider the ensemble $\mathcal{E} = \{\pi_j, \rho_j\}_{j=1,\dots,m}$ with $\pi_j = \frac{1}{m}$, $\rho_j = |\psi_j\rangle\langle\psi_j|$, where $|\psi_j\rangle$ are given by (29) with $r_0 < r_1$ $(b < 0)$. The conjectured optimal observable $\mathcal{M}$ [2] has $m + \frac{m(m-1)}{2} = \frac{m(m+1)}{2}$ components of the form:

$$\begin{aligned} M_k &= t^{-2}|\tilde{e}_k\rangle\langle\tilde{e}_k|; \quad k = 1, \dots, m; \\ M_{rs} &= \frac{2}{m}\left(1 - t^{-2}\right)|rs\rangle\langle rs|; \quad 1 \leq r < s \leq m, \end{aligned} \tag{55}$$

where $t \geq 1$, the vectors $|\tilde{e}_k\rangle$ are given by (38), and

$$|rs\rangle = \frac{1}{\sqrt{2mr_1}}\left(|\psi_r\rangle - |\psi_s\rangle\right) = \frac{1}{\sqrt{2}}\left(|e_r\rangle - |e_s\rangle\right).$$

Therefore the dual ensemble $\mathcal{E}'$ consists of the states (we use the same notation as for similar objects in the acute case):

$$\sigma_k = p_k^{-1}|\tilde{\phi}_k\rangle\langle\tilde{\phi}_k|\,, \quad p_k = \langle\tilde{\phi}_k|\tilde{\phi}_k\rangle,$$

where

$$\begin{aligned} |\tilde{\phi}_k\rangle &= t^{-1}\bar{\rho}^{1/2}|\tilde{e}_k\rangle = t^{-1}\left(\sqrt{r_1}|e_k\rangle + \frac{\sqrt{r_0}t - \sqrt{r_1}}{\sqrt{m}}|e_0\rangle\right), \\ p_k &= t^{-2}\left(r_1 + \frac{r_0 t^2 - r_1}{m}\right); \quad k = 1, \dots, m, \end{aligned}$$

and

$$\sigma_{rs} = p_{rs}^{-1}|\tilde{\phi}_{rs}\rangle\langle\tilde{\phi}_{rs}|\,, \quad p_{rs} = \langle\tilde{\phi}_{rs}|\tilde{\phi}_{rs}\rangle,$$

where

$$\begin{aligned} |\tilde{\phi}_{rs}\rangle &= \sqrt{\frac{2}{m}\left(1-t^{-2}\right)}\,\overline{\rho}^{1/2}|rs\rangle = \sqrt{\frac{r_1}{m}\left(1-t^{-2}\right)}\,\left(|e_r\rangle - |e_s\rangle\right); \\ p_{rs} &= \frac{2r_1}{m}\left(1-t^{-2}\right). \end{aligned}$$

To check the optimality condition (ii.b) we proceed in parallel with the acute case, and compute

$$K\left(\sigma_k\right)|\tilde{\phi}_k\rangle = t^{-1}\left(A(t)|e_k\rangle + B(t)\sqrt{m}|e_0\rangle\right); \quad k = 1, \ldots, m,$$

where $A(t), B(t)$ are given by (43), (44). Also

$$K\left(\sigma_{rs}\right)|\tilde{\phi}_{rs}\rangle = -\sum_j |e_j\rangle\langle e_j|\tilde{\phi}_{rs}\rangle\,\log p_{rs}^{-1}\,\left|\langle e_j|\tilde{\phi}_{rs}\rangle\right|^2, \tag{56}$$

with

$$\begin{aligned} \langle e_j|\tilde{\phi}_{rs}\rangle &= \sqrt{\frac{r_1}{m}\left(1-t^{-2}\right)}\left(\delta_{jr} - \delta_{js}\right), \\ \left|\langle e_j|\tilde{\phi}_{rs}\rangle\right|^2 &= \frac{r_1}{m}\left(1-t^{-2}\right)\left(\delta_{jr} + \delta_{js}\right). \end{aligned}$$

Substituting this into (56) results in

$$K\left(\sigma_{rs}\right)|\tilde{\phi}_{rs}\rangle = |\tilde{\phi}_{rs}\rangle, \quad 1 \le r < s \le m. \tag{57}$$

Looking again for $\Lambda_0 = c_1 I + c_0|e_0\rangle\langle e_0|$, and comparing coefficients in the condition (ii.b): $\Lambda_0|\tilde{\phi}_k\rangle = K\left(\sigma_k\right)|\tilde{\phi}_k\rangle$, we obtain the first two equations of the system (46), namely

$$\begin{aligned} c_1\sqrt{r_1} &= A(t), \\ \left(c_0 + c_1\right)\sqrt{r_0}t - c_1\sqrt{r_1} &= mB(t), \end{aligned} \tag{58}$$

while $\Lambda_0|\tilde{\phi}_{rs}\rangle = K\left(\sigma_{rs}\right)|\tilde{\phi}_{rs}\rangle$ via (57) amounts to

$$\left(1-t^{-2}\right)\left(c_1 - 1\right) = 0, \tag{59}$$

since $\langle e_0|\tilde{\phi}_{rs}\rangle \equiv 0$.

Again, two cases are possible: $t = 1$ and $t > 1$. In the first case the third equation disappears while the first two lead to the same formulas (47) as in the acute case, however with $a = \sqrt{p}$, but $b = -\sqrt{\frac{1-p}{m-1}}$. Then, denoting $z_j = \langle e_j|\psi\rangle$, the condition (ii.a) brings us to the inequality:

For complex z_j satisfying $\sum_{j=1}^{m} |z_j|^2 = 1$, it holds

$$-\sum_{j=1}^{m} |z_j|^2 \log |z_j|^2 \geq \tilde{\mu}_0(p) \left|\sum_{j=1}^{m} z_j\right|^2 - \tilde{\mu}_1(p), \tag{60}$$

where

$$\tilde{\mu}_0(p) = \frac{\sqrt{\frac{p(1-p)}{m-1}} \left(\log p - \log \frac{1-p}{m-1}\right)}{\left((m-1)\sqrt{\frac{1-p}{m-1}} - \sqrt{p}\right)\left(\sqrt{p} + \sqrt{\frac{1-p}{m-1}}\right)}, \tag{61}$$

$$\tilde{\mu}_1(p) = \frac{\sqrt{p}\log p + \sqrt{\frac{1-p}{m-1}} \log \frac{1-p}{m-1}}{\left(\sqrt{p} + \sqrt{\frac{1-p}{m-1}}\right)}. \tag{62}$$

The equality in (60) is attained for $z_1 = \sqrt{p}$, $z_j = -\sqrt{\frac{1-p}{m-1}}$, $j = 2, \ldots, m$, and all the permutations and total change of phase of these z_j.

Note that here $\tilde{\mu}_0(p) \leq 0$, which prevents us from passing to the probabilities $t_j = |z_j|^2$. However, as the lemma 1 shows, we can always restrict to the case of real z_j.

In (60) the parameter p varies in the range $[\max\left\{\frac{m-1}{m}, p(m)\right\}, 1]$, where $p(m)$ is found below. The accessible information in this case is given by the same expression (41) as in the case of moderately acute pyramids. For future use we note that

$$-\tilde{\mu}_1\left(\frac{m-1}{m}\right) = \log m - \frac{m-2}{m}\log(m-1). \tag{63}$$

If $t > 1$, then (59) imply $c_1 = 1$ and equations (58) reduce to

$$\begin{aligned} \sqrt{r_1} &= A(t), \\ c_0 &= \frac{m}{\sqrt{r_0}t}B(t) + \frac{1}{t}\sqrt{\frac{r_1}{r_0}} - 1. \end{aligned} \tag{64}$$

Here the first equation serves for determination of t (compatible with the condition $t > 1$). By using (43) it can be transformed to

$$\log \frac{m\,(m-1+\tau^2)}{2} - \frac{m-1+\tau}{m} \log{(m-1+\tau)}^2 - \frac{(1-\tau)}{m} \log{(1-\tau)}^2 = 0, \tag{65}$$

where $\tau = \sqrt{\frac{r_0}{r_1}}t$.

Denoting $g(\tau)$ the function in equation (65), we have

$$g(0) = \log \frac{m}{2} - \frac{m-2}{m} \log{(m-1)} \begin{cases} > 0, & m \leq 6; \\ < 0, & m \geq 7, \end{cases} .$$

The function is decreasing in the interval $[0, 1]$ and $g(1) = -\log 2 < 0$, hence it has a root $\tau_o(m)$ in this interval if and only if $m \leq 6$. For $\tau > 1$ the function is negative and has no roots.

The condition $t > 1$ amounts to $\tau_o(m) > \sqrt{\frac{r_0}{r_1}}$, which via (30), is equivalent to

$$r_0 < \left(1 + \frac{m-1}{\tau_o(m)^2}\right)^{-1} \equiv r_0(m),$$

and, via (32) and (51), to

$$\begin{aligned} p \equiv a^2 \quad &< \quad \left[1 + (m-1)\left(1 - \frac{m}{1-\tau_o(m)}\right)^{-2}\right]^{-1} \\ &= \quad \frac{(m-1+\tau_o(m))^2}{m(m-1+\tau_o(m)^2)} \equiv p(m). \end{aligned} \tag{66}$$

One can check that equation (65) with $\tau = \tau_o(m)$ is equivalent to

$$-\tilde{\mu}_1(p(m)) = 1. \tag{67}$$

The numerically found values[3]:

m	2	3	4	5	6	7
$\tau_o(m)$	0	0,3616	0,25866	0,15195	0,04781	[-0,05286]
$mr_0(m)$	0	0,1841	0,08726	0,02869	0,00274	–
$p(m)$	0,5	0,8725	0,8656	0,85699	0,84896	0,8417
$\frac{m-1}{m}$	0,5	0,6(6)	0,75	0,8	0.83(3)	0,8571

[3]By using equation (65) it is possible to express the function $p(m)$ via the Lambert W-function.

Here $mr_0(m)$ show a good agreement with the values found empirically for $m \leq 6$ in [2], eq. (39). In the case $m = 7$, as well as for greater values of m, the solution of the equation (65) becomes negative and also $p(m) < \frac{m-1}{m}$. This makes plausible the hypothesis that for $m \geq 7$ the optimal observable is SRM, and the corresponding entropy inequality is (60) for all $p \geq \frac{m-1}{m}$.

Then, for $m \leq 6$, under the condition (66) the equation (59) implies $c_1 = 1$ while from (64)

$$c_0 = \left(\tau_o(m)^{-1} - 1\right) \log \frac{2\left(\tau_o(m) - 1\right)^2}{m\left(m - 1 + \tau_o(m)^2\right)}.$$

The condition (ii.a) then amounts to the entropy inequality (60) with $\tilde{\mu}_0 = c_0/m, \quad \tilde{\mu}_1 = -1$ i.e.

$$-\sum_{j=1}^{m} |z_j|^2 \log |z_j|^2 \geq \frac{\left(\tau_o(m)^{-1} - 1\right)}{m} \log \frac{2\left(\tau_o(m) - 1\right)^2}{m\left(m - 1 + \tau_o(m)^2\right)} \left|\sum_{j=1}^{m} z_j\right|^2 + 1. \tag{68}$$

Due to lemma 1 one can restrict here to real z_j , $\sum_{j=1}^{m} |z_j|^2 = 1$. The equality is attained for $z_1 = -z_2 = 1/\sqrt{2}, z_j = 0,\ j \geq 3$; for $z_1 = \sqrt{p(m)}$, $z_j = -\sqrt{\frac{1-p(m)}{m-1}}$, $j \geq 2$; and for all permutations and total change of sign of these z_j.

In this case the accessible information, by using (52), is

$$\begin{aligned} A(\mathcal{E}) &= \log \frac{m}{2} - r_0 \left(\tau_o(m)^{-1} - 1\right) \log \frac{2\left(\tau_o(m) - 1\right)^2}{m\left(m - 1 + \tau_o(m)^2\right)} \\ &= \log \frac{m}{2} - r_0 m \tilde{\mu}_0(p(m)). \end{aligned} \tag{69}$$

The following theorem summarizes the content of this section.

Theorem 3. *If $m \geq 7$ then $p(m) < \frac{m-1}{m}$ and the entropy inequality (60) holds for $p \in \left[\frac{m-1}{m}, 1\right]$. If $m \leq 6$ then $p(m) > \frac{m-1}{m}$ and the inequality (60) holds for $p \in [p(m), 1]$, yielding the inequality (68) for $p \in \left[\frac{m-1}{m}, p(m)\right]$.*

The suggestion for a proof of the inequalities combining analytical argument with computer-aided reasoning is given in section 5.2.

Corollary 2. *In the case of obtuse pyramids the information-optimal observable is given by the relations (55) with $t = 1$ for $m \geq 7$, and $t = \sqrt{\frac{r_1}{r_0}}\tau_o(m)$ for $m \leq 6$. The accessible information is given by the expression (41) if $p \in \left[\max\left\{\frac{m-1}{m}, p(m)\right\}, 1\right]$, otherwise by (69) if $m \leq 6$ and $p \in \left[\frac{m-1}{m}, p(m)\right]$.*

4.5 The case of flat pyramid

The considerations above formally do not include the case of flat pyramid with $r_0 = 0, r_1 = 1/(m-1)$. Then the edges (29) of the pyramid

$$|\psi_j\rangle = \sqrt{\frac{m-1}{m}}|e_j\rangle - \sqrt{\frac{1}{m}}|e_0\rangle; \quad j = 1, \ldots, m$$

all lie in the hyperplane $\mathcal{H}_0 = \{\psi : \langle e_0|\psi\rangle = 0\}$. The vectors are linearly dependent and their convex hull is a regular symplex in $\mathcal{H}_0$. The average state of the ensemble $\mathcal{E}$

$$\overline{\rho} = \frac{1}{m-1}\left(I - |e_0\rangle\langle e_0|\right)$$

is degenerate and we should use generalized inverse in the formula (42), resulting in

$$M_j' = |\tilde{\psi}_j\rangle\langle\tilde{\psi}_j|, \quad j = 1, \ldots, m,$$

where

$$|\tilde{\psi}_j\rangle = |e_j\rangle - \sqrt{\frac{1}{m}}|e_0\rangle = \sqrt{\frac{m-1}{m}}|\psi_j\rangle. \tag{70}$$

Denoting $z_j = \langle e_j|\psi\rangle$ we have

$$\sum_{j=1}^{m} |z_j|^2 = 1, \quad \sum_{j=1}^{m} z_j = 0. \tag{71}$$

Indeed, the unit vector $|\psi\rangle$ lies in the hyperplane $\langle e_0|\psi\rangle = 0$i equality follows.

Arguing as in the previous section, we suggest that the optimal observable is given by (55) with $t = 1$ for $m \geq 7$ and $t = \infty$ for $m \leq 6$. The corresponding tight entropy inequality is

$$-\sum_{j=1}^{m} |z_j|^2 \log |z_j|^2 \geq \begin{cases} 1, & m \leq 6; \\ \log m - \frac{m-2}{m}\log(m-1), & m \geq 7, \end{cases} \tag{72}$$

for complex z_j satisfying (71) (due to a simple modification of lemma 1 one can restrict here to real z_j). Indeed, for $m \leq 6$ we can use the argument leading to inequality (68) with $\sum_{j=1}^{m} z_j = 0$ and $p = p(m)$. Then (67) implies the first line in the right-hand side of (72). On the other hand, for $m \geq 7$ we should use the inequality (60) with $\sum_{j=1}^{m} z_j = 0$ and $p = \frac{m-1}{m}$. Then the first term in the righthand side vanish, while $-\tilde{\mu}_1(p)$ gets the value (63) i.e. the second line in the righthand side of (72), which is greater than 1 for $m \leq 6$, but less for $m \geq 7$. We thus arrive at the following special case of theorem 3:

Proposition 2. *The tight entropy inequality (72) holds for real (or complex) z_j satisfying the conditions (71).*

In the case $m \leq 6$ the equality in (72) is attained for $z_1 = -z_2 = 1/\sqrt{2}$, $z_j = 0$ for $j \geq 3$; in the case $m \geq 7$ – for $z_1 = \sqrt{\frac{m-1}{m}}$, $z_j = -\sqrt{\frac{1}{(m-1)m}}$, $j \geq 2$, (and for all permutations and total change of phase of such z_j).

Theorem 1 then implies

Corollary 3. *For the ensemble of flat pyramid, information-optimal observable is given by (55) with $t = 1$ for $m \geq 7$ and $t = \infty$ for $m \leq 6$.*

The corresponding accessible information is

$$A(\mathcal{E}) = \log m - \mathrm{Tr}\,\Lambda_0 \bar{\rho} = \begin{cases} \log \frac{m}{2}, & m \leq 6; \\ \frac{m-2}{m} \log(m-1), & m \geq 7. \end{cases}$$

For $m = 3$ the inequality (72) reduces to (54). Indeed, as shown in lemma 1, one can restrict to real z_j in (72). Then $|\psi\rangle$ is a vector in the real plane $\mathcal{H}_0$ and by (70)

$$z_j = \langle e_j|\psi\rangle = \langle \tilde{\psi}_j|\psi\rangle = \sqrt{2/3}\langle \psi_j|\psi\rangle = \sqrt{2/3}\cos\alpha_j, \quad j = 1, 2, 3,$$

where α_j is the angle between the vector $|\psi\rangle$ and the ensemble vector $|\psi_j\rangle$ in the plane $\mathcal{H}_0$. The angle between any two different vectors of the ensemble is $2\pi/3$ because $\langle \psi_j|\psi_k\rangle = -1/2$ according to (34). Therefore $\alpha_j = \alpha + \frac{2\pi j}{3}$, $\quad j = 1, 2, 3$, thus we come to (54).

Let us also explain how (54) follows from the result of the paper [14]. A simple calculation shows that (54) is equivalent to

$$\log 3 - \frac{1}{3}\sum_{j=1}^{3}\left(1 + \cos\left(2\alpha + \frac{4\pi j}{3}\right)\right)\log\left(1 + \cos\left(2\alpha + \frac{4\pi j}{3}\right)\right) \geq \log 2;$$

A special case of lemma 3 in [14] (with $M = 3$) implies that the deductible in the lefthand side is maximized for $\alpha = \pi/2 + \pi k$ (giving in fact the accessible information $\log(3/2)$). The proof in [14] is based on the decomposition of the function $(1 + x)\log(1 + x)$ into the power series and a clever manipulation with the terms of the decomposition.

5 Suggestions for proofs

5.1 The case of theorem 2

Let $m \geq 2,\ \dfrac{m-1}{m} \leq p < 1$ and let $\mu_0(p), \mu_1(p)$ be given by (24), (25).

For $p \in [(m-1)/m, 1)$ and a probability distribution $(t_1, ..., t_m)$, it holds

$$F(t) \equiv H(t) - \Big(\mu_0(p)\Big(\sum_{k=1}^{m} \sqrt{t_k}\Big)^2 - \mu_1(p)\Big) \geq 0, \tag{73}$$

where $H(t) = -\sum\limits_{k=1}^{m} t_k \log t_k$ is the Shannon entropy.

In the special case $p = 1$, the inequality (73) is trivially satisfied by substituting $\mu_0(1) \equiv 0,\ \mu_1(1) = 0$.

Proof. The function $F(t)$ cannot reach a minimum on the boundary of the simplex of probability distributions, since the function

$$u(s) = -\sum_{i=1}^{m} (t_{*,i} + s a_i) \log(t_{*,i} + s a_i) - \mu_0(p)\Big(\sum_{i=1}^{m} \sqrt{t_{*,i} + s a_i}\Big) + \mu_1(p)$$

for a boundary point t_* and a transversal direction a has the derivative $-\infty$ at zero:

$$u'(0) = \lim_{t \to t_*} \left[-\sum_{i \in I_0} a_i \log t_i - C\Big(\sum_{i \in I_0} \frac{a_i}{\sqrt{t_i}}\Big) + O(1) \right],$$

where $I_0 = \{1 \leq j \leq m;\ t_{*,j} = 0\}$, $C > 0$ is a constant. In this case, $a_i > 0,\ i \in I_0$ (direction inside the simplex), hence $u'(0) = -\infty$. $\square$

Thus it is sufficient to check the inequality (73) only at the critical points.

Lemma 2. *The critical points $t_* = (t_{*,1}, ..., t_{*,m})$ of the function $F(t)$ have the form*

$$t_* = \Big(q/k, ..., q/k, (1-q)/(m-k), ..., (1-q)/(m-k)\Big) \tag{74}$$

up to permutations, for some integer k, $1 \leq k \leq m$ and $q \in [0, 1]$ (for $k = m$ it is assumed that $q = 1/m$). One of the critical points is $(p, (1-p)/(m-1), ..., (1-p)/(m-1))$, at which the value of the function $F(t)$ is equal to 0.

Proof. Consider the function $u(s) = F(t + sa)$, where $t = (t_1, ..., t_m)$, $a = (a_1, ..., a_m)$, $\sum\limits_i a_i = 0$, for a range of s such that $t + as$ defines a probability distribution. Then

$$\begin{aligned}\frac{d}{ds}u(s) &= -\sum_{i=1}^{m} a_i \log t_i - \mu_0(p)\Big(\sum_{i=1}^{m} \frac{a_i}{\sqrt{t_i}}\Big)\Big(\sum_{i=1}^{m} \sqrt{t_i}\Big) \\ &= \sum_{i=2}^{m} a_i \Bigg[-\log t_i + \log t_1 - \mu_0(p)\Big(\sum_{k=1}^{m} \sqrt{t_k}\Big)\Big(\frac{1}{\sqrt{t_i}} - \frac{1}{\sqrt{t_1}}\Big)\Bigg].\end{aligned}$$

The condition $\frac{d}{ds}u(0) = 0$ for a point $t_* = t$ and an arbitrary vector a (that is, the point t_* is critical) is satisfied if and only if for any $2 \leq i \leq m$ it holds

$$\log t_{*,1} - \log t_{*,i} = \mu_0(p)\Big(\sum_{k=1}^{m} \sqrt{t_{*,k}}\Big)\Big(\frac{1}{\sqrt{t_{*,i}}} - \frac{1}{\sqrt{t_{*,1}}}\Big).$$

Let us denote $C_1 = \mu_0(p)\Big(\sum\limits_{k=1}^{m} \sqrt{t_{*,k}}\Big), C_2 = \log t_{*,1} + \frac{C_1}{\sqrt{t_{*,1}}}$. The equation

$$C_2 - \log s = C_1/\sqrt{s}, \quad s > 0, \tag{75}$$

has at most two roots, since the derivatives of the right and the left parts with respect to s coincide only when $s = C_1^2/4$. But all the numbers $t_{*,1}, ..., t_{*,m}$ are solutions of the equation (75). Thus, among the values of $\{t_{*,i}\}$ there are no more than two different values, hence the lemma follows. $\square$

At the critical points (74) the inequality $F(t_*) \geq 0$ reduces to

$$\begin{aligned}&\mu_0(p, 1/m)\Big(\sqrt{q(k/m)} + \sqrt{(1-q)(1-k/m)}\Big)^2 - \mu_1(p, 1 - 1/m) \\ \leq\ \ & -q \log \frac{q}{k/m} - (1-q) \log \frac{1-q}{1-k/m},\end{aligned}$$

where $\mu_0(p,\alpha)$ and $\mu_1(p,\alpha)$ are defined as

$$\mu_0(p,\alpha) = \frac{\sqrt{\dfrac{p(1-p)}{\alpha(1-\alpha)}}\Big(\log\dfrac{p}{\alpha} - \log\dfrac{1-p}{1-\alpha}\Big)}{\Big(\sqrt{\alpha p} + \sqrt{(1-\alpha)(1-p)}\Big)\Big(\sqrt{\dfrac{p}{\alpha}} - \sqrt{\dfrac{1-p}{1-\alpha}}\Big)},$$

$$\mu_1(p,\alpha) = \frac{\sqrt{\dfrac{p}{\alpha}}\log\dfrac{p}{\alpha} - \sqrt{\dfrac{1-p}{1-\alpha}}\log\dfrac{1-p}{1-\alpha}}{\sqrt{\dfrac{p}{\alpha}} - \sqrt{\dfrac{1-p}{1-\alpha}}}$$

for real variables p and α varying in the range $(0,1)$.

But this inequality is a special case of lemma 4 below, the proof of which is based on the behavior of the difference between binary entropy and a function containing both a linear component and a square root term $\sqrt{t(1-t)}$. We perform a qualitative analysis of the minimizer of this function. The global minimum (there can be no more than two local minima) appears on either the interval $[0,1/2]$ or $[1/2,1]$. Importantly, this segment contains no other critical points. Therefore, minimizing the function reduces to identifying the critical point within the appropriate interval and evaluating the function at that point. A detailed proof of this statement follows from

Lemma 3. *Let $A>0,\ B,\ C\in\mathbb{R}$ (the notation of the coefficients is used only within the framework of this lemma). Then the function*

$$g(t) = h(t) - (A\sqrt{t(1-t)} + Bt + C) \tag{76}$$

reaches a local minimum on $[0,1]$ at no more than two points $t_1,t_2:\ 0<t_1\le 1/2\le t_2<1$ which are critical. In this case, if $B>0$ and if $t_*>1/2$ is critical (or $t_*<1/2$ and $B<0$), then t_* is a point of the global minimum.

Proof. We have

$$g'(0) = -\infty, \quad g'(1) = +\infty,$$

therefore local minima are on the interval $(0,1)$.

In the calculations we will pass to the natural logarithm.

$$g'(t) = -\ln\frac{t}{1-t} - \frac{A\sqrt{t(1-t)}}{2}\Big(\frac{1}{t} - \frac{1}{1-t}\Big) - B =$$
$$= -\ln\frac{t}{1-t} - \frac{A(1-2t)}{2\sqrt{t(1-t)}} - B;$$
$$g''(t) = -\frac{1}{t(1-t)} + \frac{A}{4(t(1-t))^{3/2}}.$$
$$g''(t) = 0 \quad \Leftrightarrow \quad A = 4\sqrt{t(1-t)}.$$

Therefore, there are two inflection points located on different sides of $t = 1/2$. It follows that there are three critical points $t_1 \leq t_0 \leq t_2$, where $t_1,\ t_2$ are local minimum points.

The last statement obviously follows from the symmetric case. Indeed, for $B < 0$, it holds $g(t) \leq g(1-t)$ for $t \in [1/2, 1]$, therefore the global minimum lies on the segment $[1/2, 1]$. Let $s,\ 1-s$ $(0 < s < 1/2)$ be zeros of $g''(t)$. On the segment $[s, 1-s]$ the function $g(t)'$ is decreasing, on the segments $[0, s],\ [1-s, 1]$ the function $g(t)'$ increases, and due to symmetry $(g(t) + Bt)'$ has root $1/2$. Therefore, on the segment $[1/2, 1]$ when $B < 0$ the function $g(t)$ has no more than one critical point. □

Let us finally proceed to the proof of lemma 4. The variables $\alpha,\ p,\ \beta,\ t$ used in what follows, unless otherwise stated, are such that

$$\alpha \in (0, 1/2], \quad p \in [1-\alpha, 1), \quad \beta \in [\alpha, 1/2], \quad t \in [0, 1].$$

Lemma 4. *For $\alpha,\ p,\ \beta,\ t$ specified above one has the inequality*

$$\mu_0(p,\alpha)\Big(\sqrt{\beta t} + \sqrt{(1-\beta)(1-t)}\Big)^2 - \mu_1(p,\alpha)$$
$$\leq \quad -t\log\frac{t}{\beta} - (1-t)\log\frac{1-t}{1-\beta}.$$

Proof. Let

$$f(t) = -t\log\frac{t}{\beta} - (1-t)\log\frac{1-t}{1-\beta} - \Big[\mu_0(p,\alpha)\Big(\sqrt{\beta t} + \sqrt{(1-\beta)(1-t)}\Big)^2 - \mu_1(p,\alpha)\Big],$$

for which $A = 2\mu_0(p,\alpha)\sqrt{\beta(1-\beta)}$, $B = \log\dfrac{1-\beta}{\beta} + \mu_0(p,\alpha)(2\beta - 1)$ in the form (76) of lemma 3.

The derivative of $f(t)$ is

$$f'(t) = \log\frac{1-t}{t} + \log\frac{\beta}{1-\beta} - \mu_0(p,\alpha)\left(\sqrt{\frac{\beta}{t}} - \sqrt{\frac{1-\beta}{1-t}}\right)\left(\sqrt{\beta t} + \sqrt{(1-\beta)(1-t)}\right).$$

Consider the following cases:

1. $\underline{B \leq 0.}$

   In this case there is minimum point $t_* = \beta$ ($f'(\beta) = 0$) of the function $f(t)$ on $[0, 1/2]$ due to lemma 3 presented later. One has

   $$f(\beta) = \mu_1(p,\alpha) - \mu_0(p,\alpha) \geq 0. \tag{77}$$

   Indeed, the inequality (77) is equivalent to

   $$\left(\sqrt{\frac{p}{\alpha}}\log\frac{p}{\alpha} - \sqrt{\frac{1-p}{1-\alpha}}\log\frac{1-p}{1-\alpha}\right)\left(\sqrt{p\alpha} + \sqrt{(1-p)(1-\alpha)}\right) - \\ - \sqrt{\frac{p(1-p)}{\alpha(1-\alpha)}}\left(\log\frac{p}{\alpha} - \log\frac{1-p}{1-\alpha}\right) \geq 0 \quad \Leftrightarrow$$

   $$p\log\frac{p}{\alpha}\left(1 - \sqrt{\frac{\alpha(1-p)}{p(1-\alpha)}}\right) - (1-p)\log\frac{1-p}{1-\alpha}\left(1 - \sqrt{\frac{p(1-\alpha)}{\alpha(1-p)}}\right) \geq 0 \quad \Leftrightarrow$$

   $$\sqrt{\alpha p}\log\frac{p}{\alpha} + \sqrt{(1-\alpha)(1-p)}\log\frac{1-p}{1-\alpha} \geq 0. \tag{78}$$

   Let $V(p) = \sqrt{\alpha p}\ln\frac{p}{\alpha} + \sqrt{(1-\alpha)(1-p)}\ln\frac{1-p}{1-\alpha}$, then

   (a)

   $$(d/dp)^2 V(p) = -\frac{\sqrt{\alpha}\ln(p/\alpha)}{4p^{3/2}} - \frac{\sqrt{1-\alpha}\ln((1-p)/(1-\alpha))}{4(1-p)^{3/2}} \geq 0$$

   (since $p \geq 1-\alpha \geq 1/2$);

   (b) $V(1-\alpha) = 0$ and

   $$V'(1-\alpha) = \sqrt{\frac{\alpha}{1-\alpha}} - 1 + \frac{1}{2}\ln\frac{\alpha}{1-\alpha} + \frac{1}{2}\sqrt{\frac{\alpha}{1-\alpha}}\ln\frac{1-\alpha}{\alpha} \geq 0 \quad \Leftrightarrow$$

   $$\frac{\alpha}{1-\alpha}\left(1 + \frac{1}{2}\ln\frac{1-\alpha}{\alpha}\right) - \left(1 - \frac{1}{2}\ln\frac{1-\alpha}{\alpha}\right) \geq 0 \quad \Leftrightarrow$$

   $$\ln\frac{1-\alpha}{\alpha} \geq 2 - 4\alpha.$$

Therefore, $V(p) \geq 0$ and the inequality (78) is established.

2. $\underline{B \geq 0.}$

   This condition implies that the function $f(t)$ has the minimum point t_* on $[1/2, 1]$ (lemma 3 is applied in the same way). This point is characterized by the equation

$$\log \frac{1-t_*}{t_*} + \log \frac{\beta}{1-\beta} - \mu_0(p,\alpha)\left(\sqrt{\frac{\beta}{t_*}} - \sqrt{\frac{1-\beta}{1-t_*}}\right)\left(\sqrt{\beta t_*} + \sqrt{(1-\beta)(1-t_*)}\right) = 0.$$

   The value of t_* depends on β, and for the function $f(\beta, t) \equiv f(t)$ it holds $\frac{d}{d\beta} f(\beta, t_*(\beta)) = \frac{\partial}{\partial\beta} f(\beta, t_*(\beta)) + \frac{dt_*(\beta)}{d\beta}\frac{\partial}{\partial t} f(\beta, t_*(\beta)) = \frac{\partial}{\partial\beta} f(\beta, t_*(\beta))$ since t_* is critical for $f(t)$. Then,

$$\frac{d}{d\beta} f(t_*) = \frac{1}{\ln 2}\left(\frac{t_*}{\beta} - \frac{1-t_*}{1-\beta}\right) - \mu_0(p,\alpha)\left(\sqrt{\frac{t_*}{\beta}} - \sqrt{\frac{1-t_*}{1-\beta}}\right) \times$$
$$\times\left(\sqrt{\beta t_*} + \sqrt{(1-\beta)(1-t_*)}\right).$$

   Hence

$$\ln 2 \, \frac{d}{d\beta} f(t_*) = \left(\frac{t_*}{\beta} - \frac{1-t_*}{1-\beta}\right) + \sqrt{\frac{t_*(1-t_*)}{\beta(1-\beta)}}\left(\ln \frac{1-t_*}{t_*} - \ln \frac{1-\beta}{\beta}\right) \geq 0$$

   since

$$\sqrt{\frac{1-\beta}{\beta}\frac{t_*}{1-t_*}} - \sqrt{\frac{\beta}{1-\beta}\frac{1-t_*}{t_*}} + \ln\left(\frac{\beta}{1-\beta}\frac{1-t_*}{t_*}\right) = R - 1/R - 2\ln R \geq 0$$

   where $R \equiv \sqrt{\frac{1-\beta}{\beta}\frac{t_*}{1-t_*}} \geq 1.$

   It is simple to calculate that for $\beta = \alpha$, we have $t_* = p$, and $f(p) = 0$. Therefore, $f(t_*) \geq 0$ for all $\alpha \leq \beta \leq 1/2$. We obtained the desired inequality $f(t) \geq 0$ for the minimizer t_* and hence for all the other points $t \in [0, 1]$.

$\square$

In the special case of (78), where $\alpha = 1/2$, one has an interesting inequality

$$h_{1/2}(p) \equiv \sqrt{2p}\log(2p) + \sqrt{2(1-p)}\log(2(1-p)) \geq 0,$$

which takes place for $p \in (0,1)$. The function $h_{1/2}(p)$ is nonnegative, convex and $h_{1/2}(p) \leq \sqrt{2}$.

5.2 The case of theorem 3

This section is devoted to proving the inequality (60) which verifies the condition (ii.a) in the case of obtuse pyramid. Namely, *if the parameter* $p < 1$ *satisfies* $p \geq \max\left\{\frac{m-1}{m}, p(m)\right\}$ *with* $p(m)$ *defined in* (66)*, then for complex numbers* $z_1, ..., z_m$ *satisfying (14) the inequality*

$$\tilde{F}(z) \equiv -\sum_{j=1}^{m} |z_j|^2 \log|z_j|^2 - \tilde{\mu}_0(p)\Big|\sum_{j=1}^{m} z_j\Big|^2 + \tilde{\mu}_1(p) \geq 0.$$

holds with $\tilde{\mu}_0(p), \tilde{\mu}_1(p)$ *defined in (61), (62).*

Suggestion for proof with computer verification. As it has been established in lemma 1, it suffices to consider real vectors.

For real vectors $(z_j)_{j=1}^m$ on the unit sphere let us consider the Lagrange function

$$L(z_1, ..., z_k) = -\sum_{k=1}^{m} z_k^2 \log z_k^2 - \tilde{\mu}_0(p)\Big(\sum_{k=1}^{m} z_k\Big)^2 + \tilde{\mu}_1(p) - \lambda\Big(\sum_{k=1}^{m} z_k^2 - 1\Big).$$

The first-order conditions yield

$$-\frac{1}{2}\frac{\partial L}{\partial z_k} = z_k \log z_k + \tilde{\mu}_0(p)\Big(\sum_{j=1}^{m} z_j\Big) + (\lambda + \log e) z_k = 0, \quad 1 \leq k \leq m,$$

and the set $\{z_j\}$ has at most three different values s_1, s_2, s_3 with multiplicities $k_1, k_2, k_3 \geq 0$ $(k_1 + k_2 + k_3 = m)$, since equation

$$z \log z^2 + \tilde{\mu}_0(p)\Big(\sum_{j=1}^{m} z_j\Big) + (\lambda + \log e) z = 0 \tag{79}$$

in a variable z has at most three real solutions. The fact that the equation of the form $z \log z^2 + az + b = 0$ for real a and b has no more than three real roots was actually used in the proof of lemma 2 for equation (75) which is similar to (79) with ≤ 2 roots. However, here we consider the entire real line, so the maximum number of real roots is three.

We consider two cases.

Case 1. The minimizer has a zero component.
Then $\lambda + \log e = 0$ and the equation (79) has roots 0 and $\pm s$. Therefore the minimizer of $\tilde{F}(z)$ must be of the form $(\underbrace{s, -s, ..., s, -s}_{k \text{ times}}, 0, ..., 0)$, where $2ks^2 = 1$ giving the value $-2ks^2 \log s^2 + \tilde{\mu}_1(p) = \log 2k + \tilde{\mu}_1(p)$ which has the minimum equal to $1 + \tilde{\mu}_1(p)$ for $k = 1$. The value $1 + \tilde{\mu}_1(p)$ is nonnegative for the following reason.

First, we prove that the function $\tilde{\mu}_1(p)$ is increasing for p in the range $\Big[\dfrac{m-1}{m}, 1\Big)$. The derivative is

$$\tilde{\mu}_1'(p) = \log e \frac{2(m-2)\sqrt{\dfrac{p(1-p)}{m-1}} - 4p + 2 + \ln p - \ln \dfrac{1-p}{m-1}}{2(m-1)\sqrt{\dfrac{p(1-p)}{m-1}}\left(\sqrt{p} + \sqrt{\dfrac{1-p}{m-1}}\right)^2}.$$

Here, $2(m-2)\sqrt{\dfrac{p(1-p)}{m-1}} \geq 0$ and $\ln \dfrac{p}{1-p} - 4p + 2 \geq 0$, hence $\tilde{\mu}_1'(p) \geq 0$.

Thus for $\frac{m-1}{m} \leq p < 1$ we have

$$-\tilde{\mu}_1(p) \leq -\tilde{\mu}_1\Big(\frac{m-1}{m}\Big) = -\frac{m-2}{m}\log(m-1) + \log m$$

by (63). For $p(m) \leq p < 1$ we have

$$-\tilde{\mu}_1(p) \leq -\tilde{\mu}_1(p(m)) = 1,$$

by (67). Therefore, $1 + \tilde{\mu}_1(p) \geq 0$ for $p > \max(p(m), (m-1)/m)$. Note that $\tilde{\mu}_1(p) \geq \tilde{\mu}_1((m-1)/m) \geq \tilde{\mu}_1(p(m)) = -1$ if $p(m) < (m-1)/m$.

Case 2. The minimizer has no zero components.
The Lagrange multipliers method predicts that the minimizer has at most 3 different values, that are in our case nonzero. They are denoted as $s_1 < 0 <$

$s_2 \le s_3$ (without loss of generality we can assume that only s_1 is negative) with multipliers k_1, k_2, k_3, such that $k_1+k_2+k_3 = m$ and $k_1 s_1^2+k_2 s_2^2+k_3 s_3^2 = 1$. Our hypothesis is that, however, there is only two distinct nonzero values among s_1, s_2, s_3. This fact can be verified by computer experiments; we performed numerical simulations for m from 3 to 50 and for the parameter $a = \tilde{\mu}_0(p) > 0$ taking values in $\{0.1, 0.5, 1, 5, 10, 100, \infty\}$ (see Appendix).

Thus, up to permutation and an overall sign change, $(z_j)_{j=1}^m = (\underbrace{s_1, ..., s_1}_{k_1 \text{ times}}, \underbrace{s_2, ..., s_2}_{k_2 \text{ times}})$, $s_1 < 0$, $s_2 > 0$. The components of the minimizer must have opposite signs; otherwise, changing the sign of a single component would decrease $\tilde{F}$.

Denote

$$\tilde{\mu}_0(p,\alpha) = \frac{\sqrt{\dfrac{p(1-p)}{\alpha(1-\alpha)}}\left(\log\dfrac{p}{\alpha} - \log\dfrac{1-p}{1-\alpha}\right)}{\left(\sqrt{(1-p)(1-\alpha)} - \sqrt{p\alpha}\right)\left(\sqrt{\dfrac{p}{\alpha}} + \sqrt{\dfrac{1-p}{1-\alpha}}\right)},$$

$$\tilde{\mu}_1(p,\alpha) = \frac{\sqrt{\dfrac{p}{\alpha}}\log\dfrac{p}{\alpha} + \sqrt{\dfrac{1-p}{1-\alpha}}\log\dfrac{1-p}{1-\alpha}}{\sqrt{\dfrac{p}{\alpha}} + \sqrt{\dfrac{1-p}{1-\alpha}}},$$

where α and p lie on $(0,1)$.

Relation with $\tilde{\mu}_0(p)$ and $\tilde{\mu}_1(p)$ is given by

$$\tilde{\mu}_0(p) = \frac{1}{m}\tilde{\mu}_0(p, 1/m),$$
$$\tilde{\mu}_1(p) = \tilde{\mu}_1(p, 1/m) - \log m.$$

It is easy to see that for α and p such that $0 < \alpha \le 1/2 \le 1-\alpha \le p < 1$ one has $\log\dfrac{p}{\alpha} - \log\dfrac{1-p}{1-\alpha} \ge 0$ and $\sqrt{(1-p)(1-\alpha)} - \sqrt{p\alpha} \le 0$, therefore, $\tilde{\mu}_0(p,\alpha) \le 0$.

The initial inequality (60) for $k_1 s_1^2 = t$ is equivalent to

$$-t\log\frac{t}{k_1/m} - (1-t)\log\frac{1-t}{k_2/m} \ge \tilde{\mu}_0(p, 1/m)\left(\sqrt{t(k_1/m)} - \sqrt{(1-t)(k_2/m)}\right)^2 - \tilde{\mu}_1(p, 1/m),$$

and the result follows from lemma 5.

$\square$

Lemma 5. *Let* $0 \le \alpha \le \beta \le \frac{1}{2} \le 1-\alpha \le p \le 1$, $0 \le t \le 1$. *Then*

$$\tilde{f}(t) \equiv -t\log\frac{t}{\beta} - (1-t)\log\frac{1-t}{1-\beta} - \left(\tilde{\mu}_0(p,\alpha)\left(\sqrt{\beta t} - \sqrt{(1-\beta)(1-t)}\right)^2 - \tilde{\mu}_1(p,\alpha)\right) \ge 0.$$

Proof. We will prove that the inequality

$$\mu_0(p,\alpha)\left(\sqrt{\beta t} + \sqrt{(1-\beta)(1-t)}\right)^2 - \mu_1(p,\alpha) \ge \tilde{\mu}_0(p,\alpha)\left(\sqrt{\beta t} - \sqrt{(1-\beta)(1-t)}\right)^2 - \tilde{\mu}_1(p,\alpha)$$

holds in this range. So, the statement of the lemma follows from lemma 4.

Collecting similar terms and using notation of $f(t)$ from subsection 5.1,

$$\tilde{f}(t) - f(t) = 2\sqrt{\beta(1-\beta)}\Big(\mu_0(p,\alpha) + \tilde{\mu}_0(p,\alpha)\Big)\sqrt{t(1-t)} - (1-2\beta)\Big(\mu_0(p,\alpha) - \tilde{\mu}_0(p,\alpha)\Big)t + \\ + \Big[(1-\beta)\Big(\mu_0(p,\alpha) - \tilde{\mu}_0(p,\alpha)\Big) - \Big(\mu_1(p,\alpha) - \tilde{\mu}_1(p,\alpha)\Big)\Big] \ge 0.$$

One can see that

$$\mu_0(p,\alpha) - \tilde{\mu}_0(p,\alpha) = 2(2p-1)\frac{\sqrt{p(1-p)\alpha(1-\alpha)}\left(\log\frac{p}{\alpha} - \log\frac{1-p}{1-\alpha}\right)}{(p-\alpha)(p+\alpha-1)},$$

$$\mu_0(p,\alpha) + \tilde{\mu}_0(p,\alpha) = -2(1-2\alpha)p(1-p)\frac{\log\frac{p}{\alpha} - \log\frac{1-p}{1-\alpha}}{(p-\alpha)(p+\alpha-1)} \le 0,$$

$$\mu_1(p,\alpha) - \tilde{\mu}_1(p,\alpha) = 2\frac{\sqrt{p(1-p)\alpha(1-\alpha)}\left(\log\frac{p}{\alpha} - \log\frac{1-p}{1-\alpha}\right)}{p-\alpha}.$$

The function $(\tilde{f} - f)(t)$ has the form $A\sqrt{t(1-t)} + Bt + C$ with $A \le 0$ and $B \le 0$, then the unique minimum point $t_* \in [1/2, 1]$. It has the value $t_* = \frac{1}{2}\left(1 - \frac{B}{\sqrt{A^2+B^2}}\right)$ and the minimum of the function $(\tilde{f} - f)(t)$ is equal to

$$(\tilde{f} - f)(t_*) = \frac{B}{2} + C - \frac{1}{2}\sqrt{A^2+B^2},$$

where

$$A = 2\sqrt{\beta(1-\beta)}\Big(\mu_0(p,\alpha) + \tilde{\mu}_0(p,\alpha)\Big),$$

$$B = -(1-2\beta)\Big(\mu_0(p,\alpha) - \tilde{\mu}_0(p,\alpha)\Big),$$

$$C = \Big[(1-\beta)\Big(\mu_0(p,\alpha) - \tilde{\mu}_0(p,\alpha)\Big) - \Big(\mu_1(p,\alpha) - \tilde{\mu}_1(p,\alpha)\Big)\Big].$$

One has

$$\frac{B}{2} + C = \frac{\sqrt{p(1-p)\alpha(1-\alpha)}\Big(\log\frac{p}{\alpha} - \log\frac{1-p}{1-\alpha}\Big)}{(p-\alpha)(p+\alpha-1)} \times$$

$$\times\Big((2p-1) - 2(p+\alpha-1)\Big) =$$

$$= (1-2\alpha)\frac{\sqrt{p(1-p)\alpha(1-\alpha)}\Big(\log\frac{p}{\alpha} - \log\frac{1-p}{1-\alpha}\Big)}{(p-\alpha)(p+\alpha-1)} \geq 0,$$

$$\frac{1}{2}\sqrt{A^2+B^2} = (1-2\alpha)\frac{\sqrt{p(1-p)\alpha(1-\alpha)}\Big(\log\frac{p}{\alpha} - \log\frac{1-p}{1-\alpha}\Big)}{(p-\alpha)(p+\alpha-1)} \times$$

$$\times\sqrt{4\,\frac{p(1-p)\beta(1-\beta)}{\alpha(1-\alpha)} + \frac{(1-2p)^2(1-2\beta)^2}{(1-2\alpha)^2}}$$

Thus,

$$\frac{\sqrt{A^2+B^2}}{B+2C} = \sqrt{4\,\frac{p(1-p)\beta(1-\beta)}{\alpha(1-\alpha)} + \frac{(1-2p)^2(1-2\beta)^2}{(1-2\alpha)^2}} \leq 1 \tag{80}$$

Let us denote $u(t) = 4t(1-t)$. The inequality (80) can be rewritten in the form

$$\frac{u(\beta)u(p)}{u(\alpha)} + \frac{(1-u(\beta))(1-u(p))}{1-u(\alpha)} \leq 1,$$

where $u(p) \leq u(\alpha) \leq u(\beta)$. The linear function

$$t \mapsto W(t) = \frac{u(\beta)t}{u(\alpha)} + \frac{(1-u(\beta))(1-t)}{1-u(\alpha)}$$

has non-negative derivative $\dfrac{u(\beta)}{u(\alpha)} - \dfrac{1-u(\beta)}{1-u(\alpha)} = \dfrac{u(\beta)-u(\alpha)}{u(\alpha)(1-u(\alpha))}$ and $W(u(\alpha)) = 1$. Therefore, the inequality (80) holds, and lemma has been proven. $\square$

Appendix. Numerical Verification of Hypotheses for Minimization on the Sphere (Generated by DeepSeek)

We consider the minimization problem of the function

$$\tilde{F}(z_1, \ldots, z_m) = -\sum_{k=1}^{m} z_k^2 \log_2(z_k^2) + a \left(\sum_{k=1}^{m} z_k \right)^2 \tag{81}$$

on the unit sphere $\sum_{k=1}^{m} z_k^2 = 1$, where the parameter $a > 0$ equals to $\tilde{\mu}_0(p)$ given by (61) in the obtuse scenario; in the flat scenario (formally corresponding to the case $a = \infty$) the second term in the righthand side of (81) is absent and there is additional constraint $\sum_{k=1}^{m} z_k = 0$.

The main hypothesis suggests that minimizers have at most two distinct nonzero values. To verify this hypothesis numerically, we implement the following verification scheme:

Algorithm 1 Numerical verification of the minimization hypothesis

Input: Parameter a, dimension range $[m_{\min}, m_{\max}]$, tolerance $\epsilon = 10^{-8}$
Output: Verification result for each dimension m
procedure VerifyHypothesis(a, $m_{\min}$, $m_{\max}$)
    for $m = m_{\min}$ **to** $m_{\max}$ **do**
        Find approximate minimizer $\mathbf{z}^*$ on the sphere
        Compute $\tilde{F}_{\min} = \tilde{F}(\mathbf{z}^*)$
        ▷ Test Hypothesis 1: Two-value structure
        $\mathbf{z}_{\text{proj}} \leftarrow$ ProjectToTwoValueStructure($\mathbf{z}^*, a$)
        if $\mathbf{z}_{\text{proj}} \neq$ None **then**
            $\Delta\tilde{F}_1 \leftarrow \tilde{F}(\mathbf{z}_{\text{proj}}) - \tilde{F}_{\min}$
            $\text{valid}_1 \leftarrow (|\Delta\tilde{F}_1| \le \epsilon)$
        else
            $\text{valid}_1 \leftarrow$ False
        end if
        ▷ Test Hypothesis 2: Special two-component vector
        $\mathbf{z}_{\text{special}} \leftarrow (1/\sqrt{2}, 0, \dots, 0, -1/\sqrt{2})$
        $\Delta\tilde{F}_2 \leftarrow \tilde{F}(\mathbf{z}_{\text{special}}) - \tilde{F}_{\min}$
        $\text{valid}_2 \leftarrow (|\Delta\tilde{F}_2| \le \epsilon)$
        ▷ Accept hypothesis if either projection is nearly optimal
        confirmed $\leftarrow \text{valid}_1$ **or** valid_2
        Output result for dimension m
    end for
end procedure

For a given vector $\mathbf{z}^*$, we compute the averages u and v of its positive and negative components, respectively, then rescale them to lie on the unit sphere. In the case $a = \infty$, the exact values $u = \sqrt{\ell/(km)}$, $v = -\sqrt{k/(\ell m)}$ are used to satisfy both $\|\mathbf{z}\| = 1$ and $\sum_i z_i = 0$. The projection replaces all positive entries by u and all negative entries by v, preserving zeros.

Verification Criterion

The hypothesis is considered confirmed for dimension m if either:

- The projection $\mathbf{z}_{\text{proj}}$ onto a two-value structure yields a function value within tolerance ϵ of the computed minimum: $|\tilde{F}(\mathbf{z}_{\text{proj}}) - \tilde{F}_{\min}| \le \epsilon$

- The special two-component vector $\mathbf{z}_{\text{special}} = (1/\sqrt{2}, 0, \ldots, 0, -1/\sqrt{2})$ yields a function value within tolerance ϵ of the computed minimum

We use $\epsilon = 10^{-8}$ throughout our computations. The optimization is performed using SciPy's SLSQP algorithm with multiple initial points to ensure robustness.

```
======================================================================
VERIFICATION OF THE MINIMIZATION HYPOTHESIS FOR FUNCTION F ON THE SPHERE
======================================================================
Enter parameter a: 10
Enter m_min: 3
Enter m_max: 50

======================================================================
Parameters: a = 10, m in [3, 50]
```

Criterion: hypothesis holds if $\Delta\tilde{F}$_1 < 1e-8 or $\Delta\tilde{F}$_2 < 1e-8

```
======================================================================
```

m	$\tilde{F}$_min	$\Delta\tilde{F}$_1	$\Delta\tilde{F}$_2	Result
3	1.00000000	2.52e-01	-9.44e-15	YES
4	1.00000000	2.08e-01	-4.11e-15	YES
5	1.00000000	1.22e-01	-4.33e-15	YES
6	0.98734793	-1.44e-15	1.27e-02	YES
7	0.91479512	0.00e+00	8.52e-02	YES
8	0.85213438	1.11e-16	1.48e-01	YES
9	0.79788958	-8.88e-16	2.02e-01	YES
10	0.75061687	1.11e-16	2.49e-01	YES
11	0.70910123	-3.33e-16	2.91e-01	YES
12	0.67236219	-3.33e-16	3.28e-01	YES
13	0.63961653	-3.33e-16	3.60e-01	YES
14	0.61023775	0.00e+00	3.90e-01	YES
15	0.58372158	-2.22e-16	4.16e-01	YES
16	0.55965888	-2.22e-16	4.40e-01	YES
17	0.53771482	-1.67e-15	4.62e-01	YES

```
 18 |  0.51761305 |    2.22e-16 |    4.82e-01 |    YES
 19 |  0.49912361 |    5.55e-17 |    5.01e-01 |    YES
 20 |  0.48205361 |   -1.11e-16 |    5.18e-01 |    YES
 21 |  0.46624010 |   -5.55e-17 |    5.34e-01 |    YES
 22 |  0.45154446 |   -7.77e-15 |    5.48e-01 |    YES
 23 |  0.43784802 |   -2.78e-16 |    5.62e-01 |    YES
 24 |  0.42504857 |   -2.50e-15 |    5.75e-01 |    YES
 25 |  0.41305761 |   -2.78e-16 |    5.87e-01 |    YES
 26 |  0.40179807 |   -8.88e-16 |    5.98e-01 |    YES
 27 |  0.39120256 |   -3.33e-16 |    6.09e-01 |    YES
 28 |  0.38121185 |   -2.22e-16 |    6.19e-01 |    YES
 29 |  0.37177368 |   -2.33e-15 |    6.28e-01 |    YES
 30 |  0.36284175 |    1.11e-16 |    6.37e-01 |    YES
 31 |  0.35437487 |    3.89e-16 |    6.46e-01 |    YES
 32 |  0.34633629 |   -2.78e-16 |    6.54e-01 |    YES
 33 |  0.33869310 |   -1.11e-16 |    6.61e-01 |    YES
 34 |  0.33141576 |   -5.55e-17 |    6.69e-01 |    YES
 35 |  0.32447761 |   -2.05e-15 |    6.76e-01 |    YES
 36 |  0.31785461 |    3.33e-16 |    6.82e-01 |    YES
 37 |  0.31152497 |   -2.66e-15 |    6.88e-01 |    YES
 38 |  0.30546889 |   -7.61e-15 |    6.95e-01 |    YES
 39 |  0.29966835 |   -3.33e-16 |    7.00e-01 |    YES
 40 |  0.29410691 |   -1.59e-14 |    7.06e-01 |    YES
 41 |  0.28876954 |   -6.11e-16 |    7.11e-01 |    YES
 42 |  0.28364247 |   -2.61e-15 |    7.16e-01 |    YES
 43 |  0.27871305 |   -1.67e-16 |    7.21e-01 |    YES
 44 |  0.27396965 |   -8.60e-15 |    7.26e-01 |    YES
 45 |  0.26940154 |   -8.60e-15 |    7.31e-01 |    YES
 46 |  0.26499883 |    3.33e-16 |    7.35e-01 |    YES
 47 |  0.26075237 |    2.78e-16 |    7.39e-01 |    YES
 48 |  0.25665367 |   -4.00e-15 |    7.43e-01 |    YES
 49 |  0.25269489 |    5.55e-16 |    7.47e-01 |    YES
 50 |  0.24886871 |   -1.67e-16 |    7.51e-01 |    YES

======================================================================
FINAL RESULTS:
======================================================================
 YES: Hypothesis CONFIRMED for all 48 values of m
```

The computational results reveal that for large a and small m (< 7), the minimizer typically contains only two nonzero entries of opposite sign. Otherwise, all components are nonzero and assume exactly two distinct values.

6 Postscriptum (added 15/07/2026)

In response to our postings `arXiv:2506.06700` (first version of this one, related to the inequalities for the Shannon entropy) and `arXiv:2603.24017` (dealing with the corresponding inequalities for the Rényi entropy in the flat case), the preprints of Haonan Zhang, `arXiv:2605.05243`, and of Alvan Arulandu, arXiv:2606.14698, appeared, suggesting independent AI-assisted proofs of our conjectured inequalities for flat and, respectively, obtuse pyramids. Further, we wish to add that inequality (28) for strongly acute pyramid is closely related to the sharp log-Sobolev inequality for simple random walk on Z_d (Pott's model) computed in P. Diaconis, L. Saloff-Coste, Logarithmic Sobolev inequalities for finite Markov chains, The Annals of Applied Probability, **6** (3), 695-750, (1996). Thus all our conjectured tight inequalities, along with their consequences for the accessible information of quantum pyramids, are completely analytically confirmed.

Here we also wish to make a clarification concerning the general optimality conditions in theorem 1. The relations **iii.a,b** of this theorem give the universally valid necessary and sufficient conditions for the optimality; however the relations **ii.a,b** are applicable, strictly speaking, only under certain additional assumption. Namely, the quantities Tr $\sigma M_j'$ in the equation (4) defining $K(\sigma)$ should be all positive in order to avoid the infinite value of the logarithms. In particular, this holds for all σ if all the operators M_j' are strictly positive. The conditions **ii.a,b** then acquire literal meaning. On the other hand, such an assumption does not hold for an ensemble of pure linearly independent states like quantum pyramids. However, while the inequality **ii.a** remains valid *formally*, tolerating the value $+\infty$ in the righthand side, the equation **ii.b** which was used to find the eigenvalues of operator Λ_0 for this ensemble, is still valid literally because the expression

$$K(\sigma_k^0)\,|\phi_k^0\rangle = -\sum_j M_j'|\phi_k^0\rangle \log\langle\phi_k^0|M_j'|\phi_k^0\rangle$$

is always well defined taking into account the usual convention $0 \log 0 = 0$. Therefore the derivation of the operator Λ_0 and of all the corresponding hypothetical entropy inequalities also remains valid. In fact, corresponding calculations could be performed on the basis of the more appropriate equation **iii.b**, which however would be less intuitive. Alternative proof of the conditions **iii.a,b** avoiding introduction of the operators $K(\sigma)$ is outlined in the paper: A.S. Holevo, Probability measures on the set of quantum states in optimization problems for information quantities, Theory Probab. Appl. **71**(2), (2026).

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